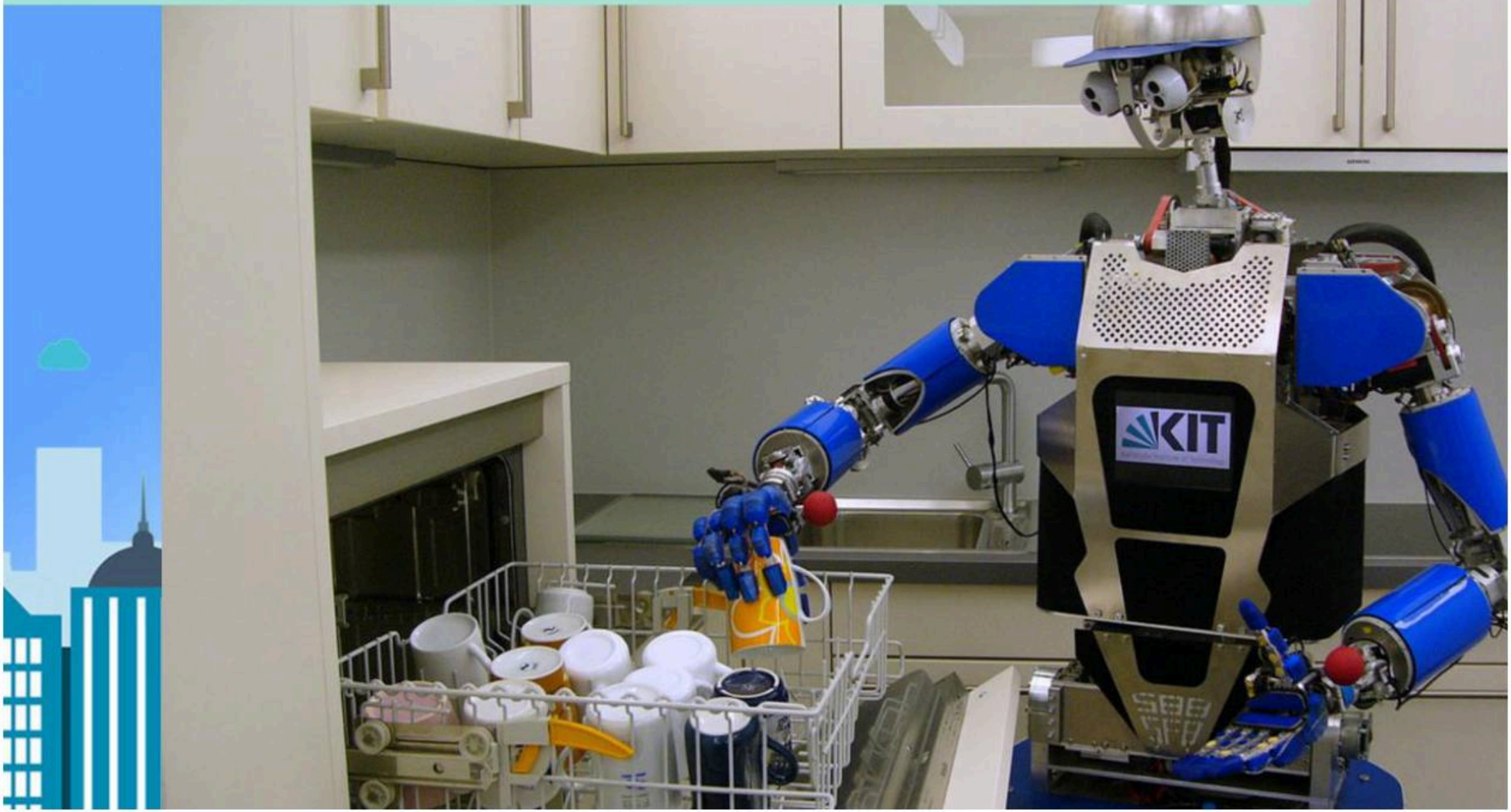


We want the robot to be able to mow the grass in the yard, remind to take medicine and deliver groceries from the shops.



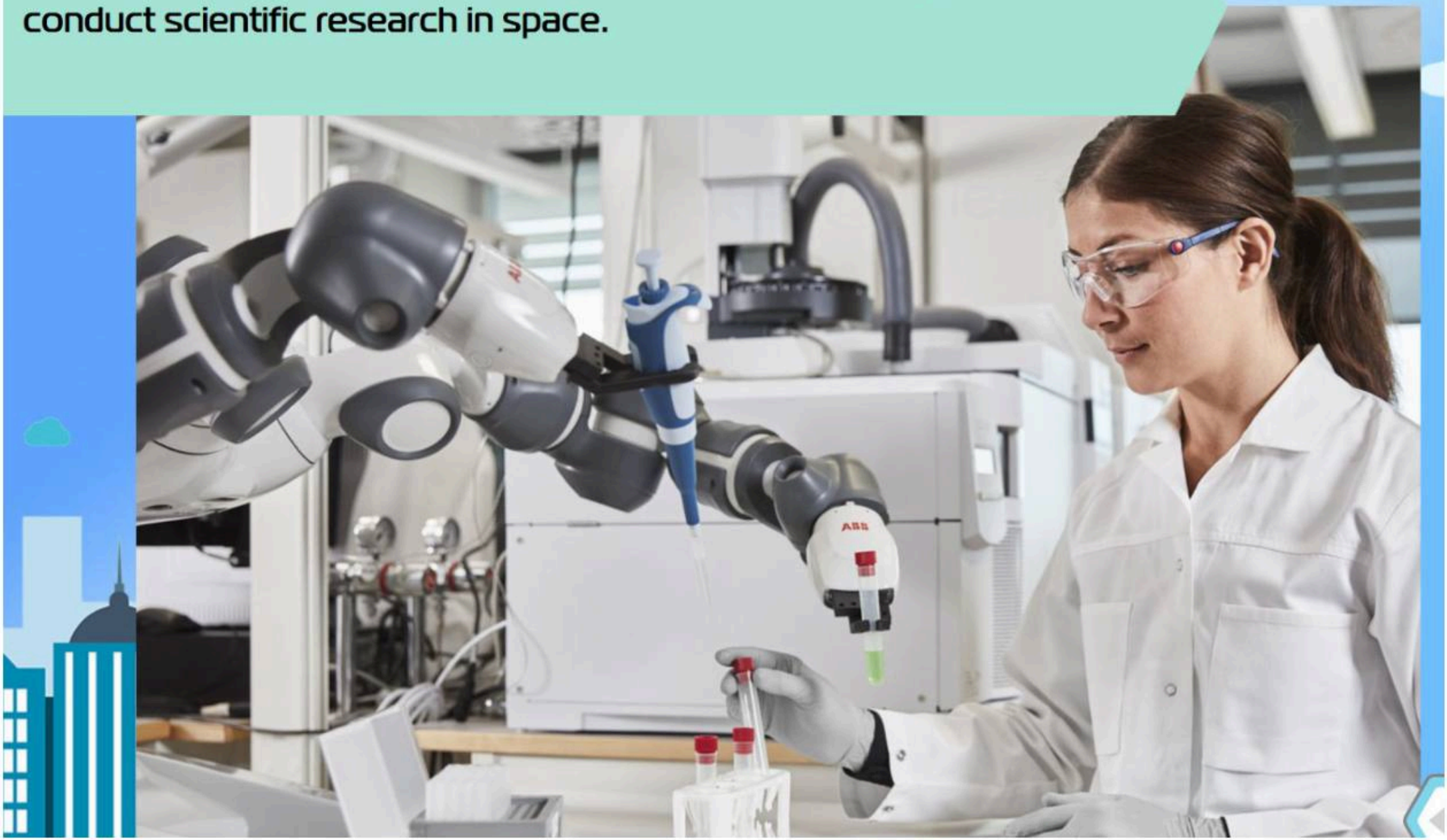


What robots do you know? What do they do?

Today we will create a home robot assistant. Recently, the heat has become simply unbearable, and we want to transfer part of the work to a robot.



Robots must work where it is dangerous - under water, put out fires, conduct scientific research in space.



Of course, different robots will be best suited for different applications.
They can look completely different.



For example, firefighting robots have a crawler chassis and are adapted to work at high temperatures, in smoke, etc.



Underwater robots currently look like boxes with manipulators. Due to the fact that they move very slowly, they do not need streamlined shapes.



And domestic robots should look like people. This construction is related to the fact that domestic robots must perform the same tasks as humans and in the same environment as humans.



So that such robots do not look threatening, they are made not tall and with cute faces



No matter how functional the robots are, it depends only on you what they will do and what you will do.



It is desirable to spend the time freed by work for the benefit of yourself and others. Watching videos can always teach you something, and not just waste time.



And instead of playing computer games, you can learn to program and first create modifications for your favorite games, and later - your own games.



This is a home robot-assistant, to which you can delegate the execution of most boring and uninteresting tasks.



Robot

Two servo motors are used in the robot chassis. SmartHub receives the commands from the controlling device and runs the motors



Find:

- Smarthub
- Left motor
- Right motor



The robot's head, and arm-shaped manipulators make the robot friendly and helpful in home chores.

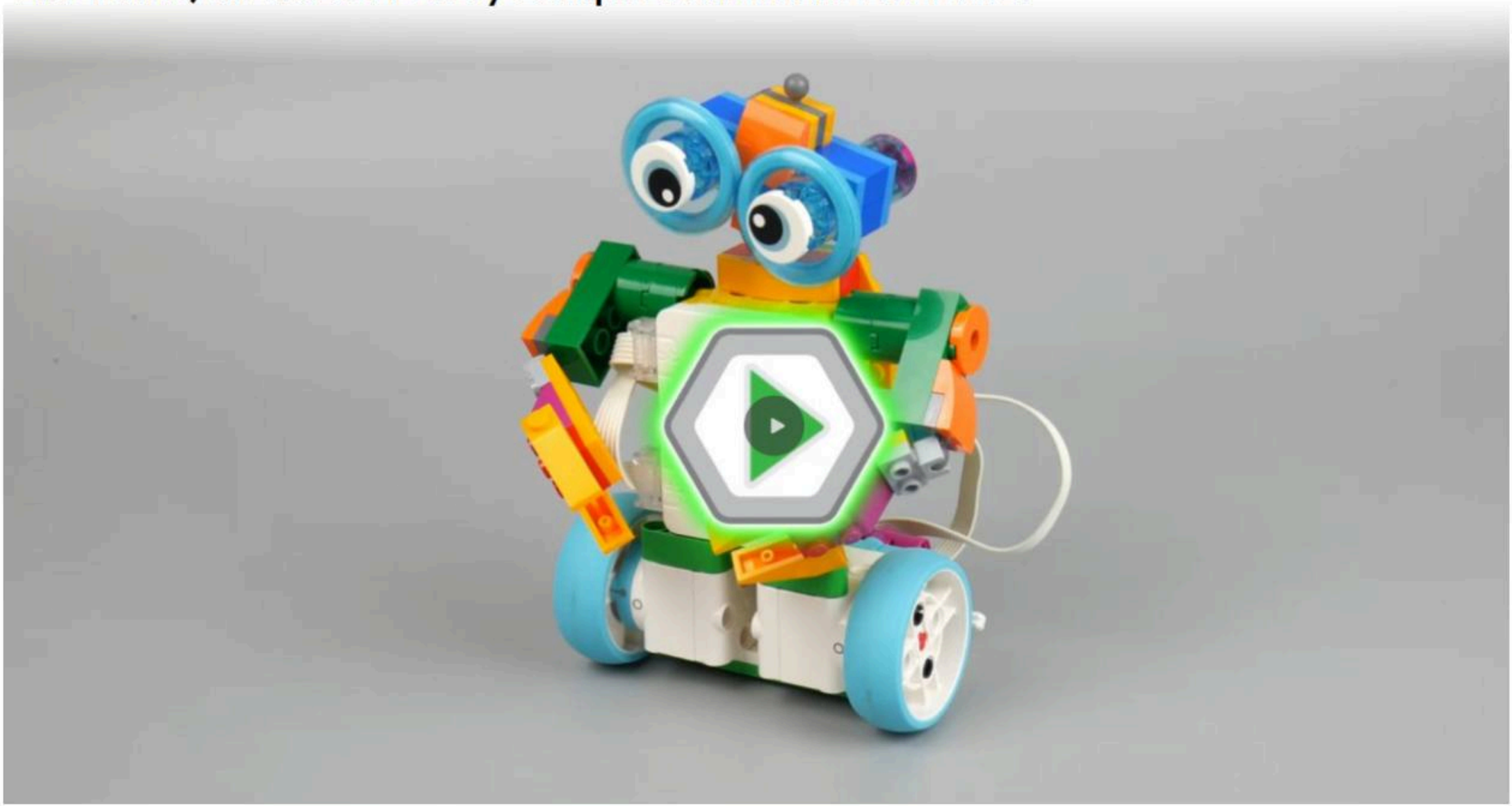


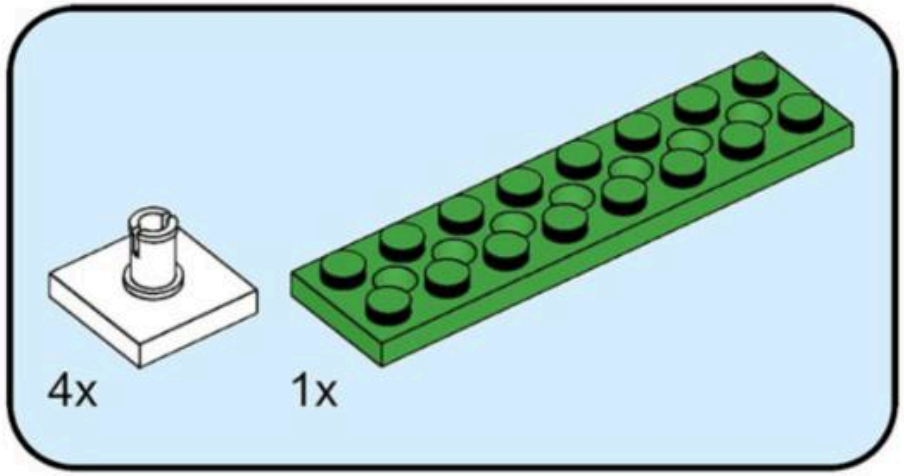
Find:

- head
- manipulators

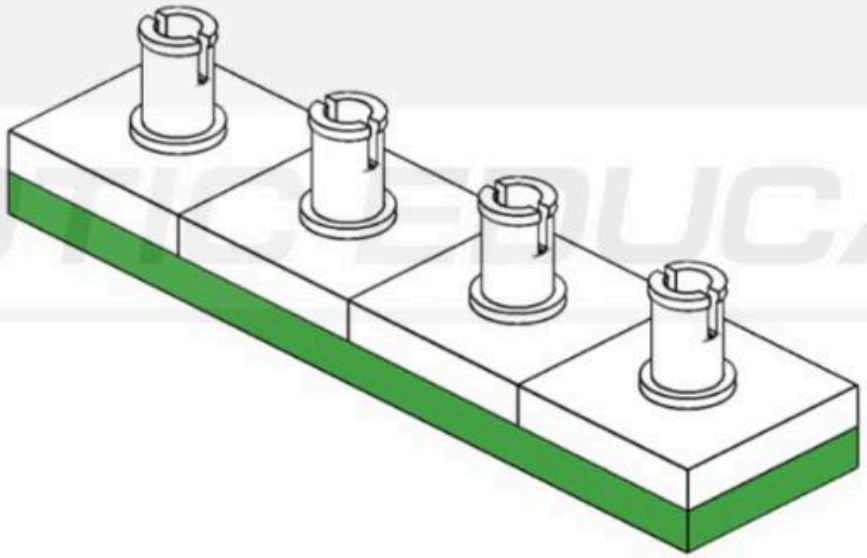


The robot has a wheeled chassis. The left and right wheels are driven by separate motors. Thanks to the independent wheel drive and the vertical placement of the smarthub, the robot is very compact and maneuverable.

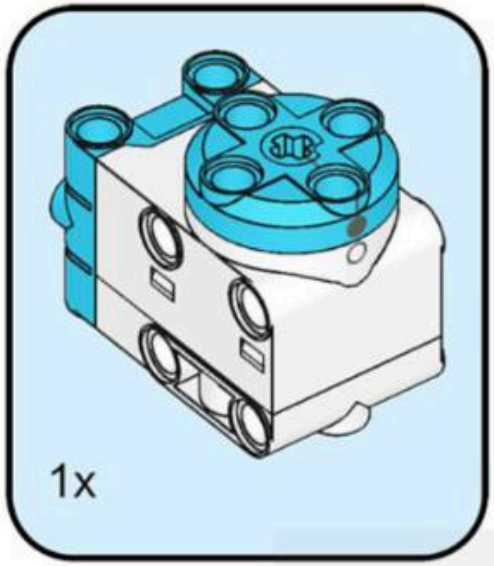




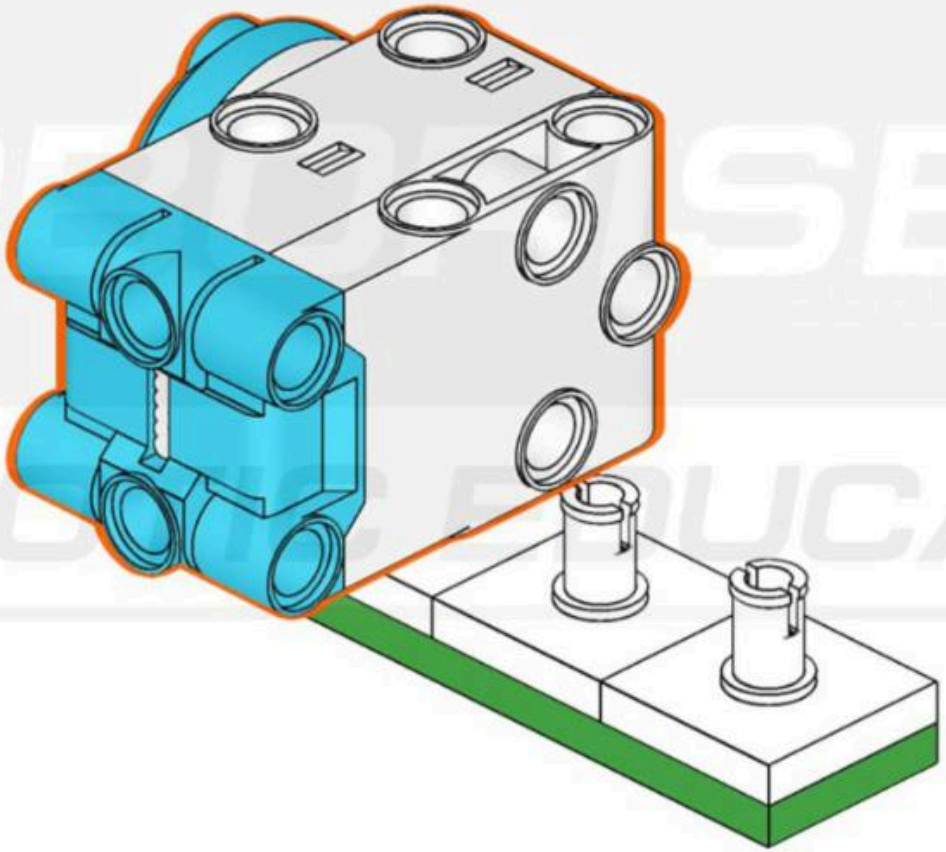
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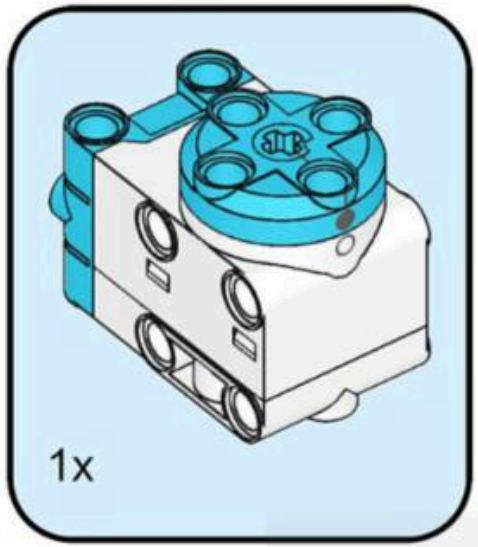


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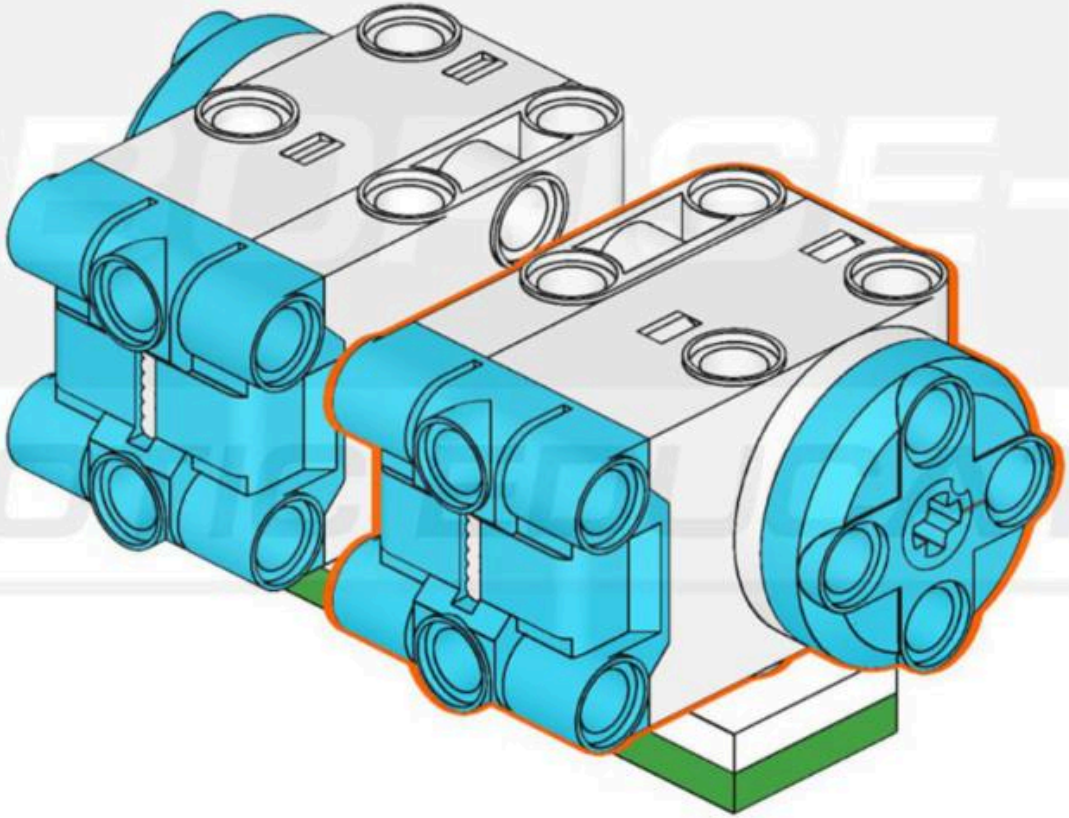


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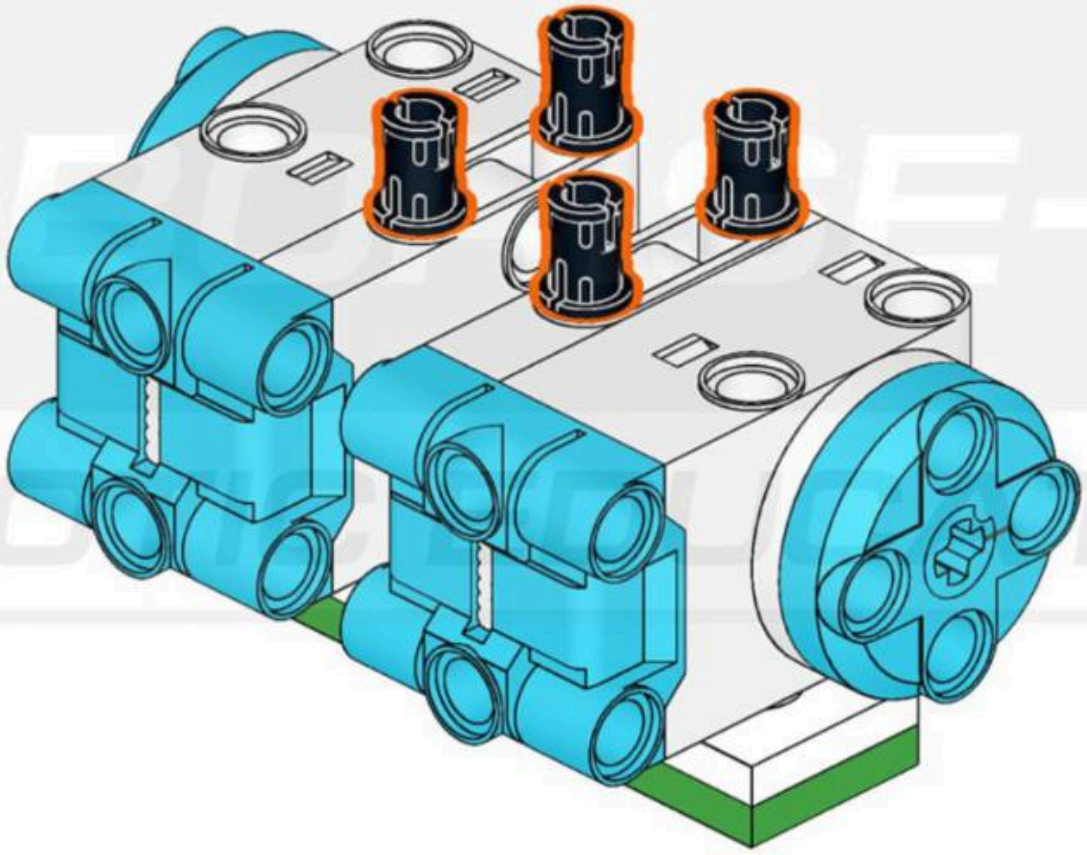


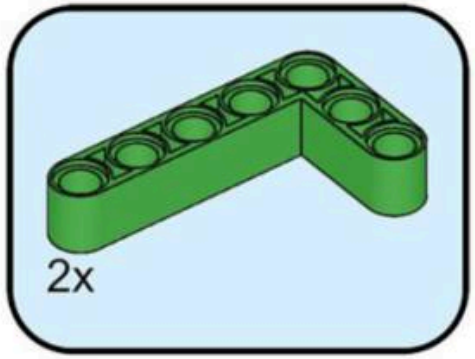
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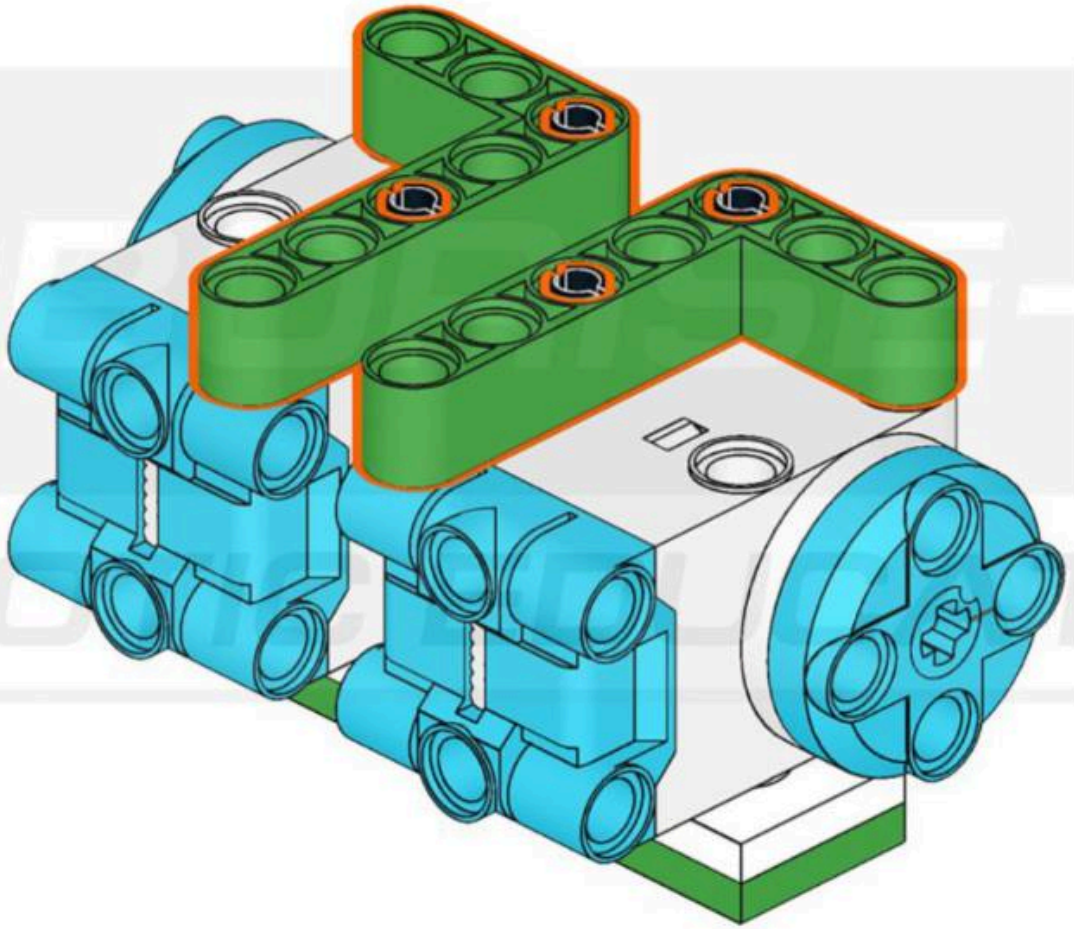


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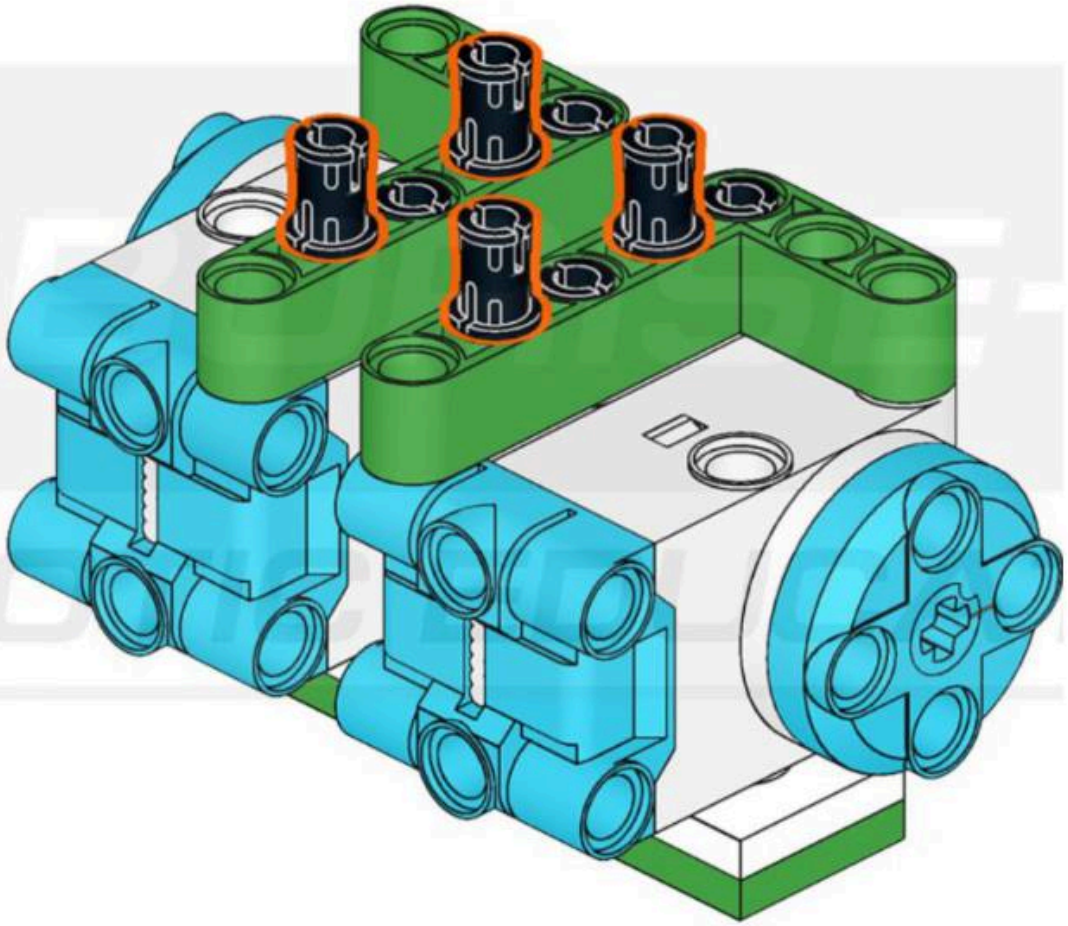
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ROBOTSHOP
ROBOTSHOP



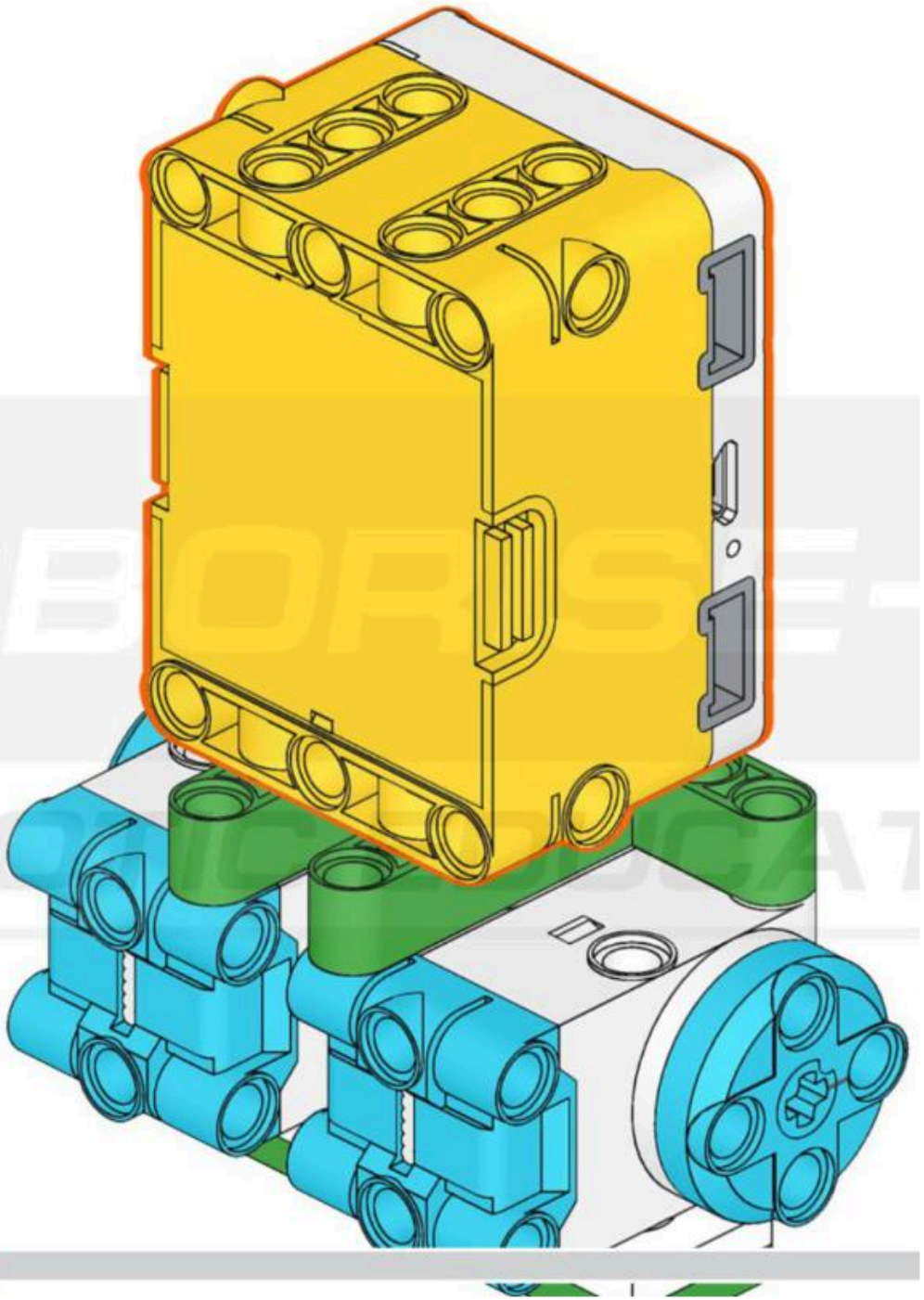
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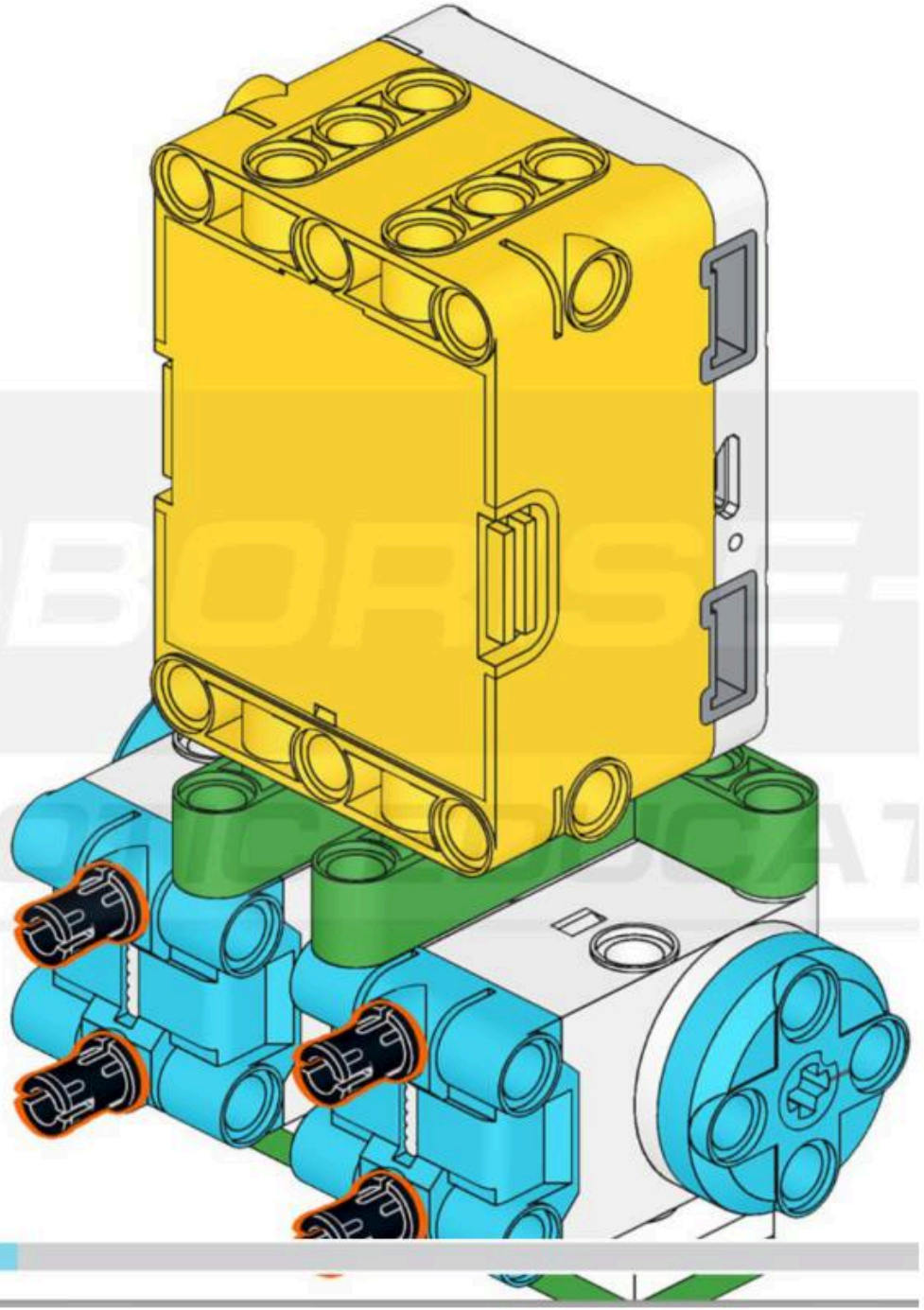


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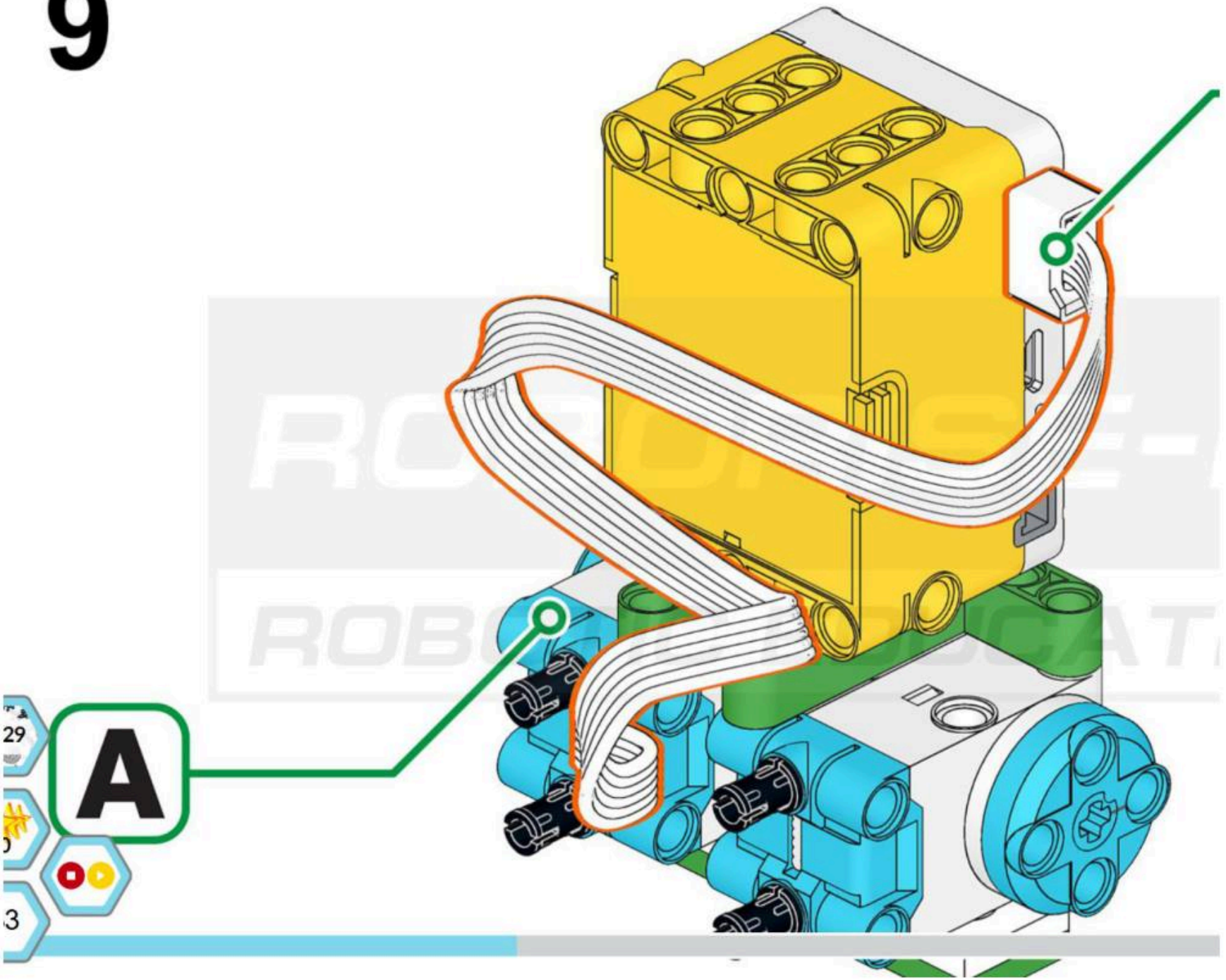


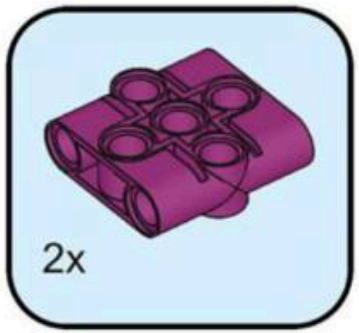
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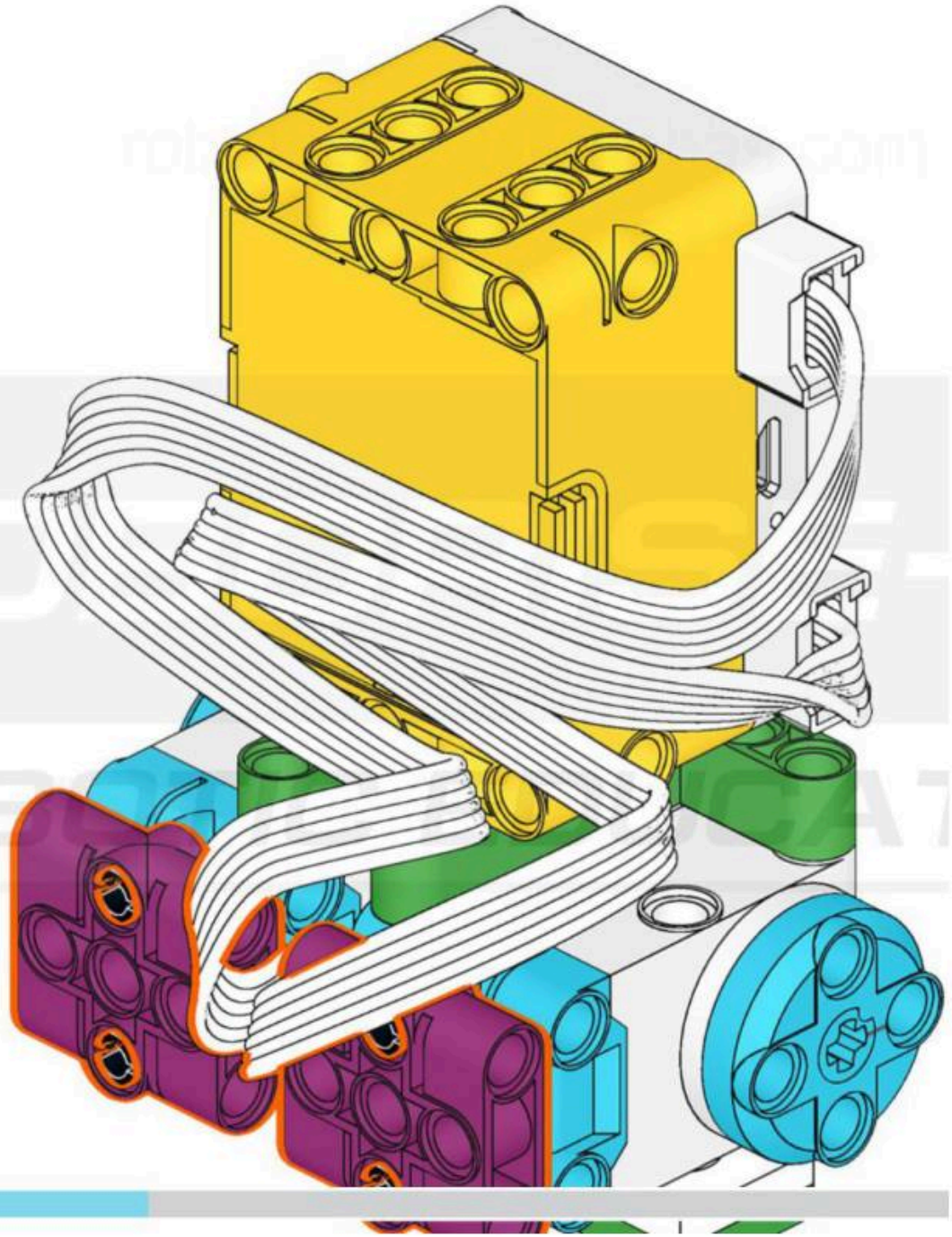
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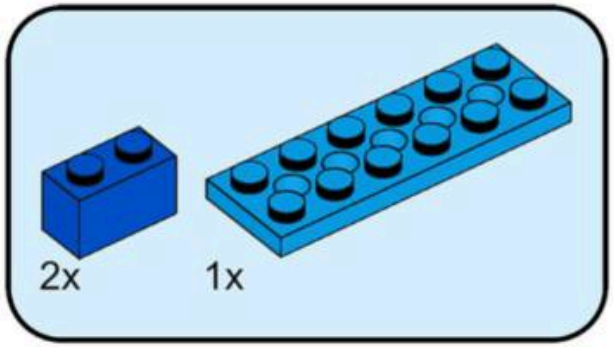
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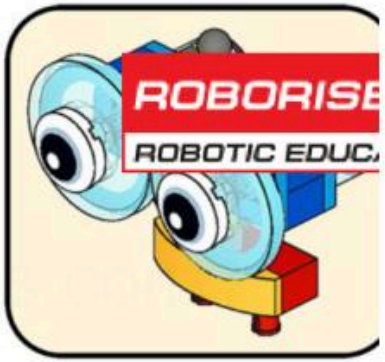




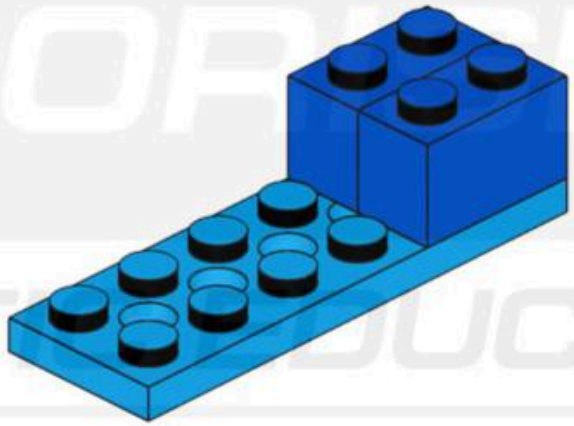
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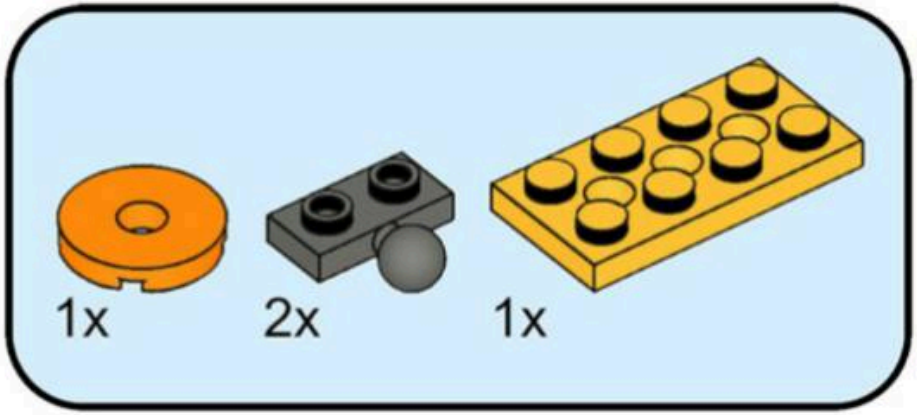




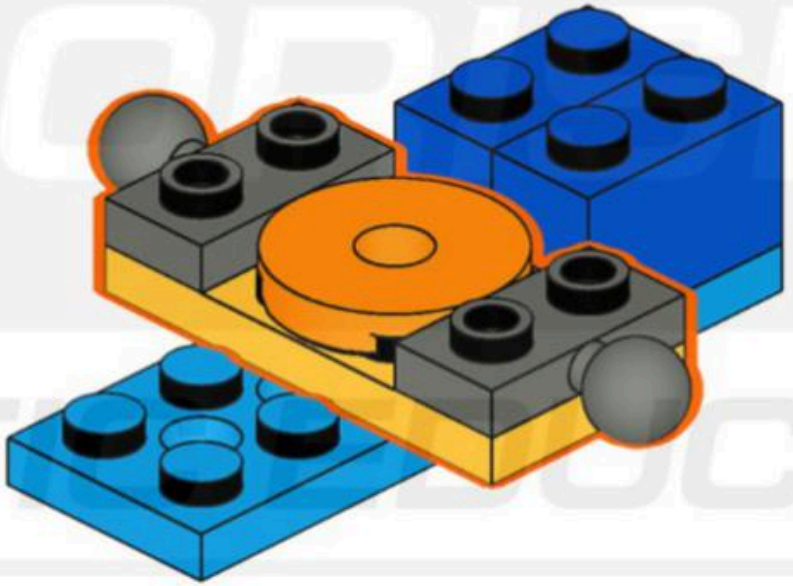
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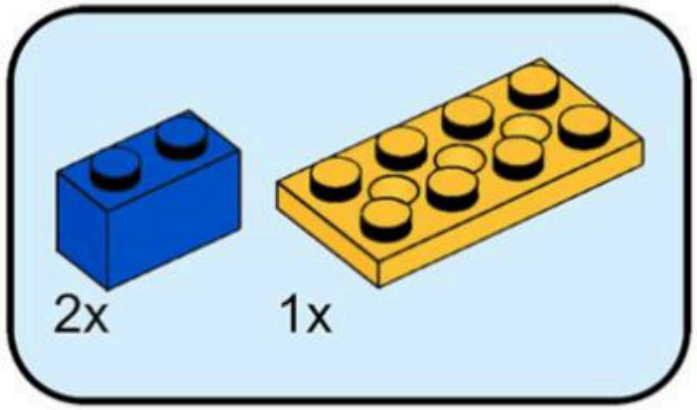


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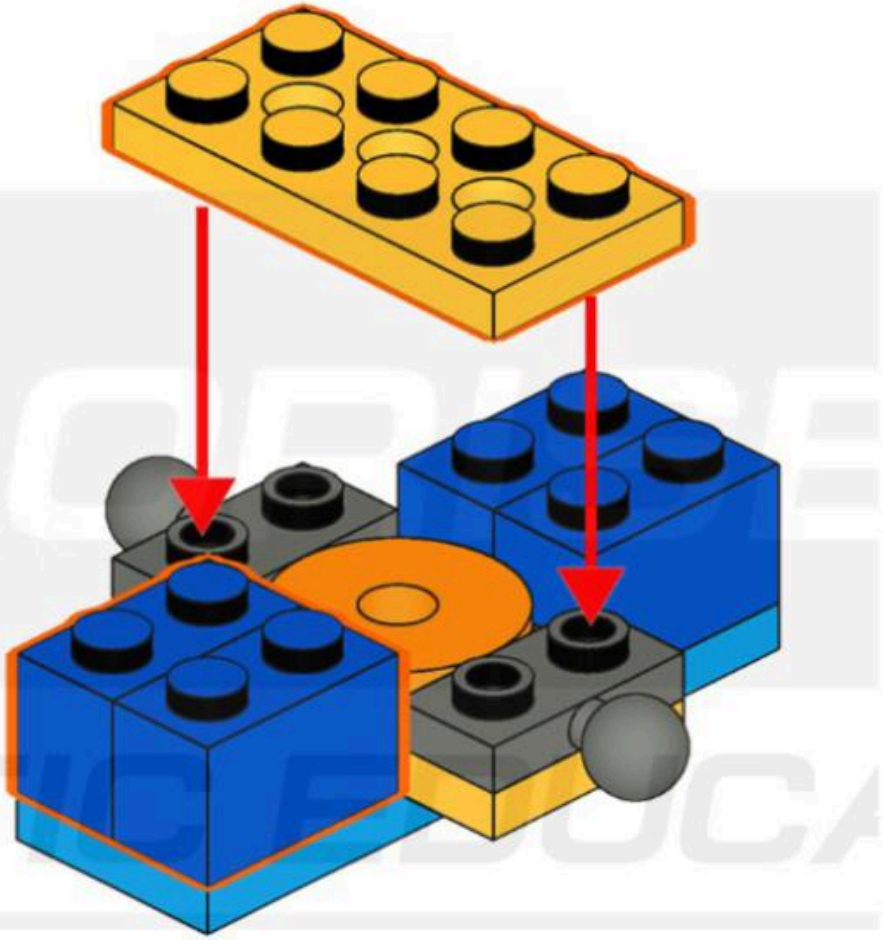


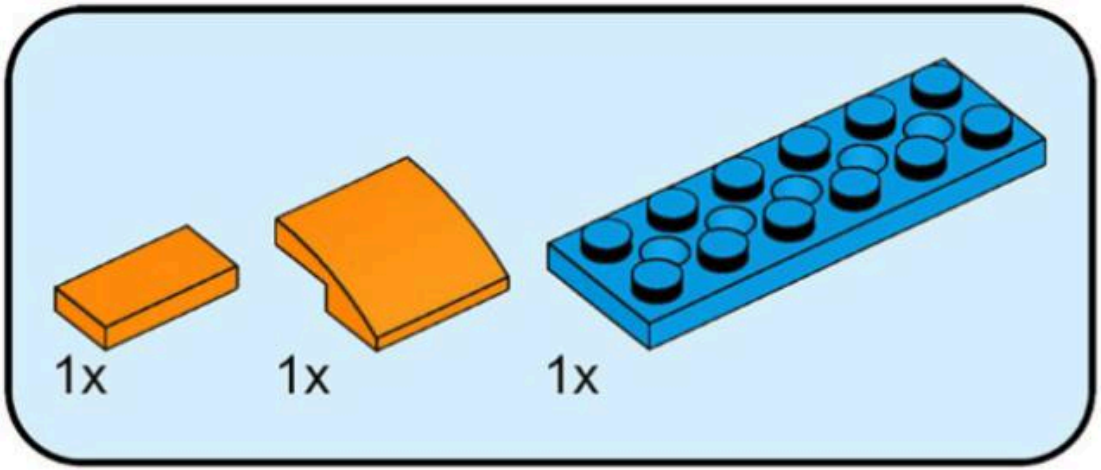
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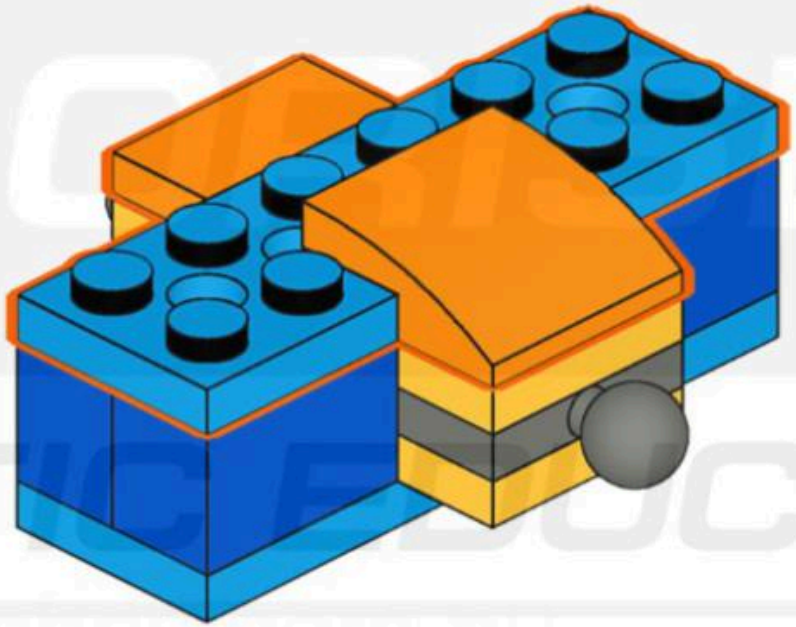


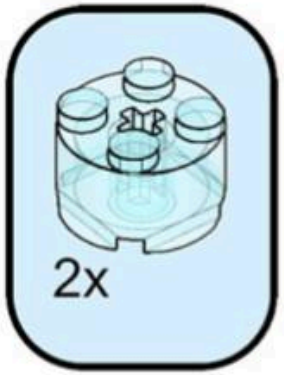
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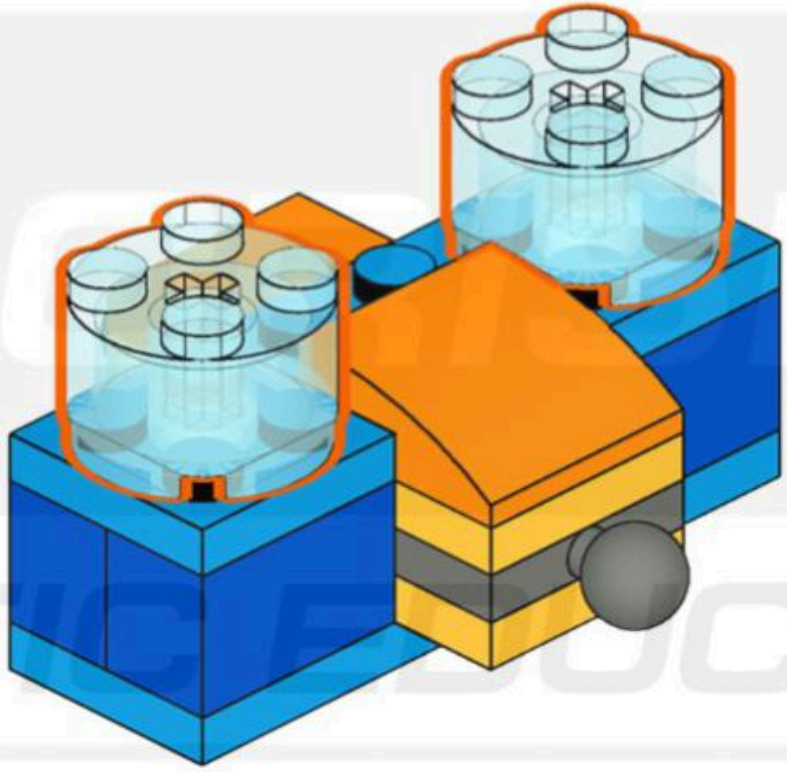


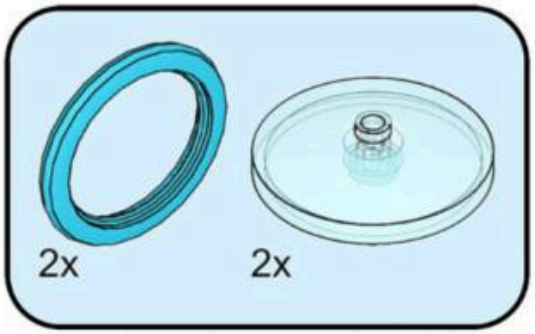
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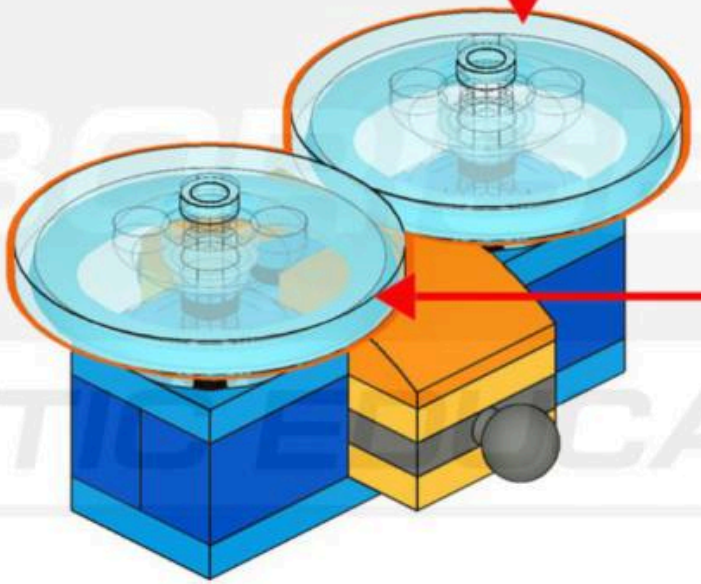


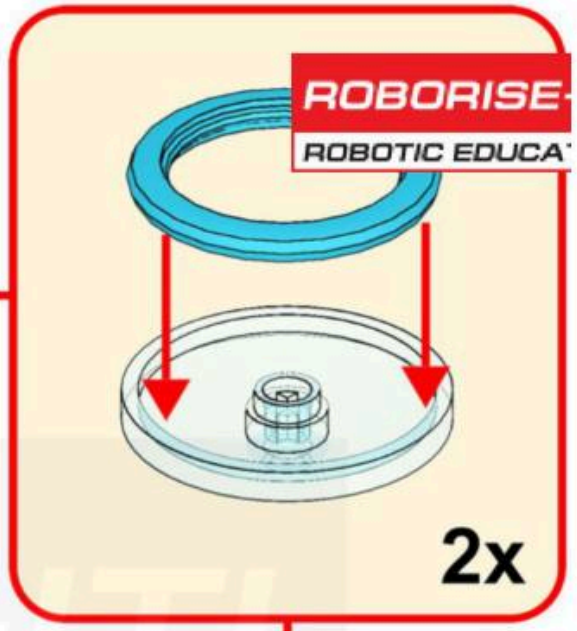
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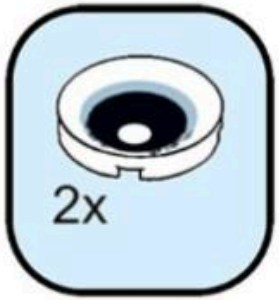


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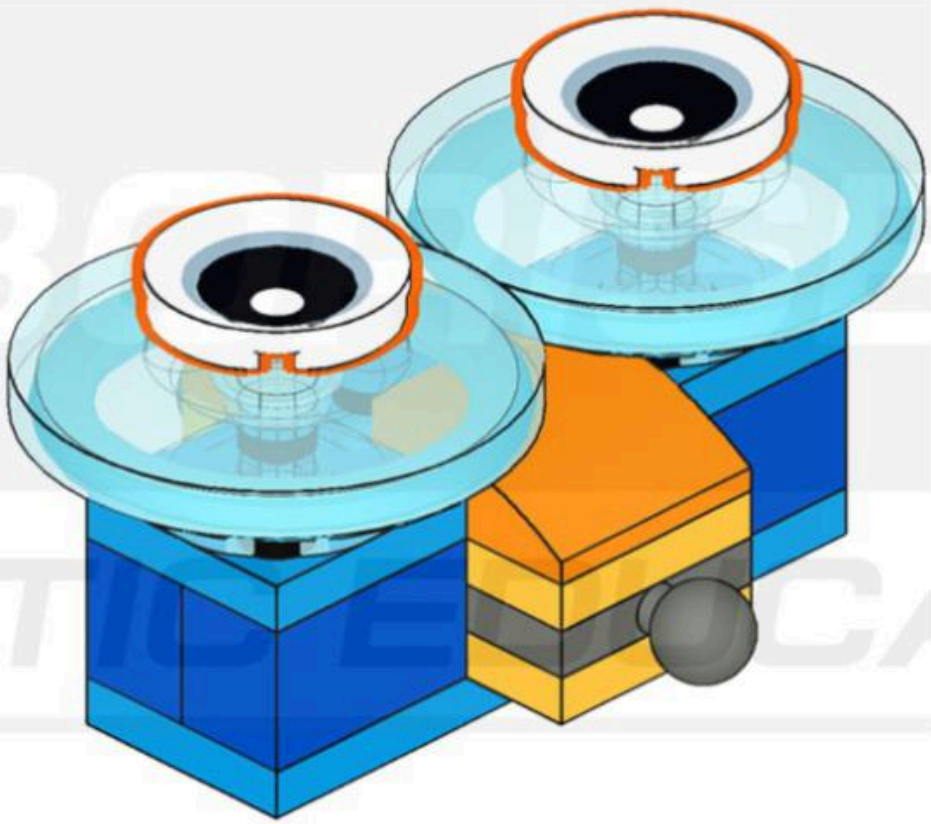


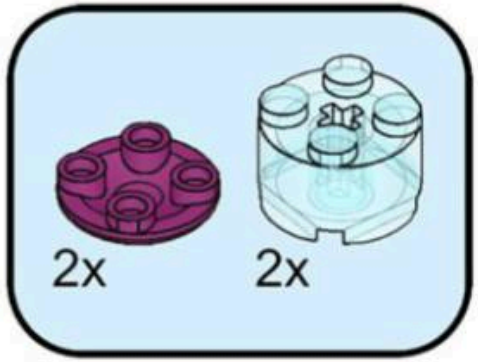


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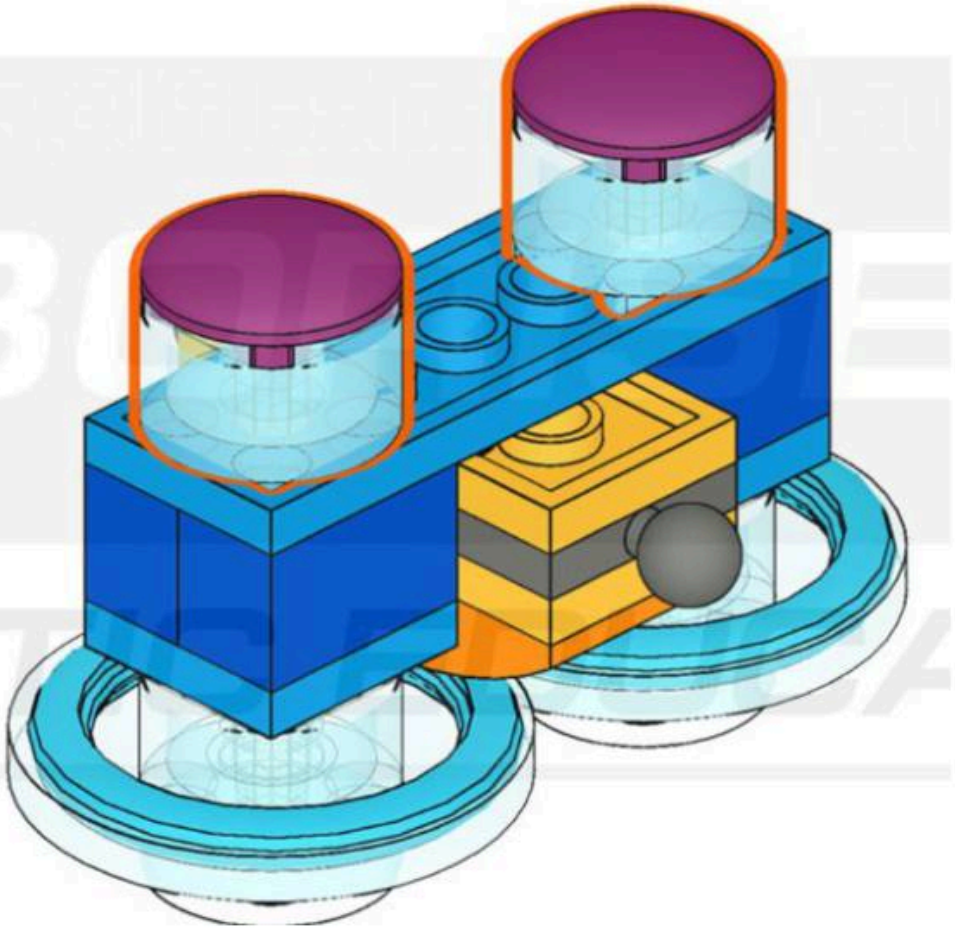


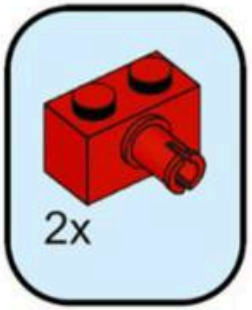
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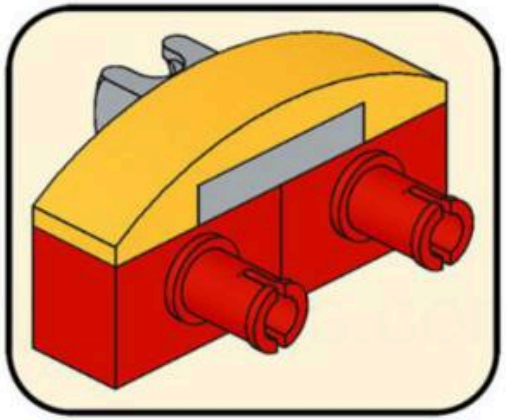


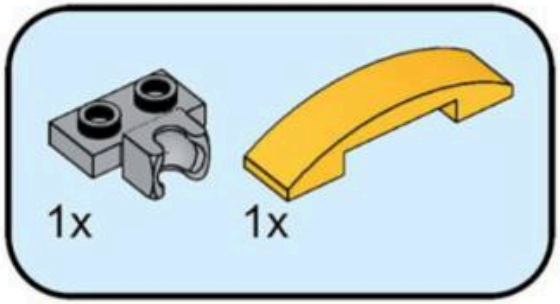
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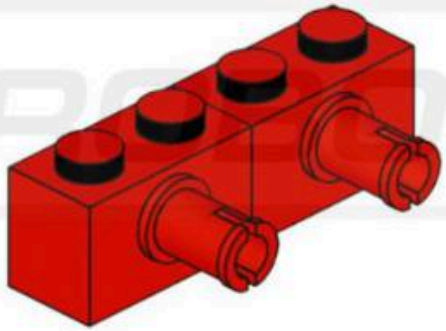


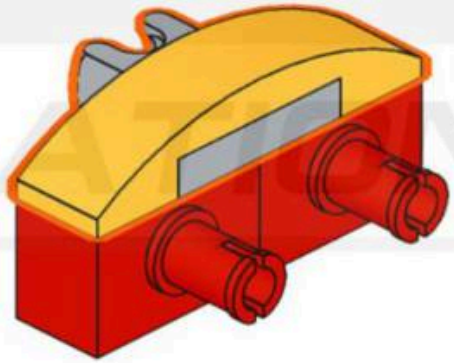
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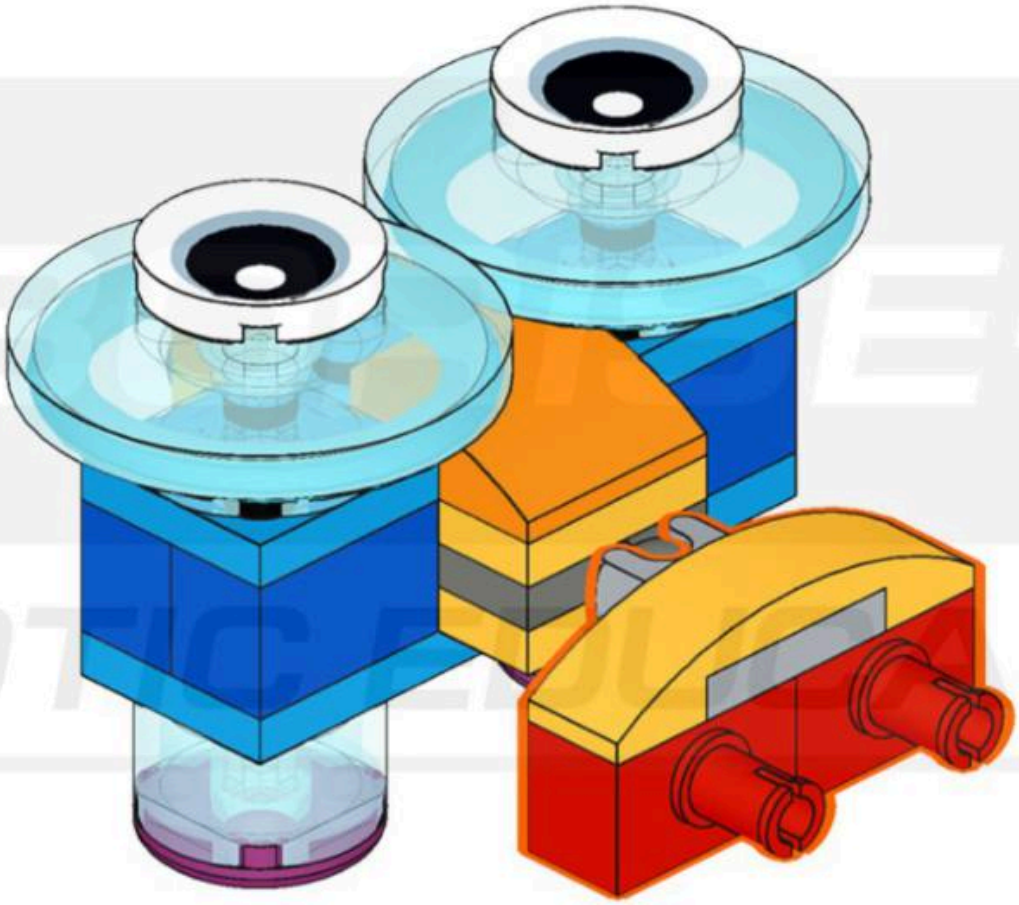
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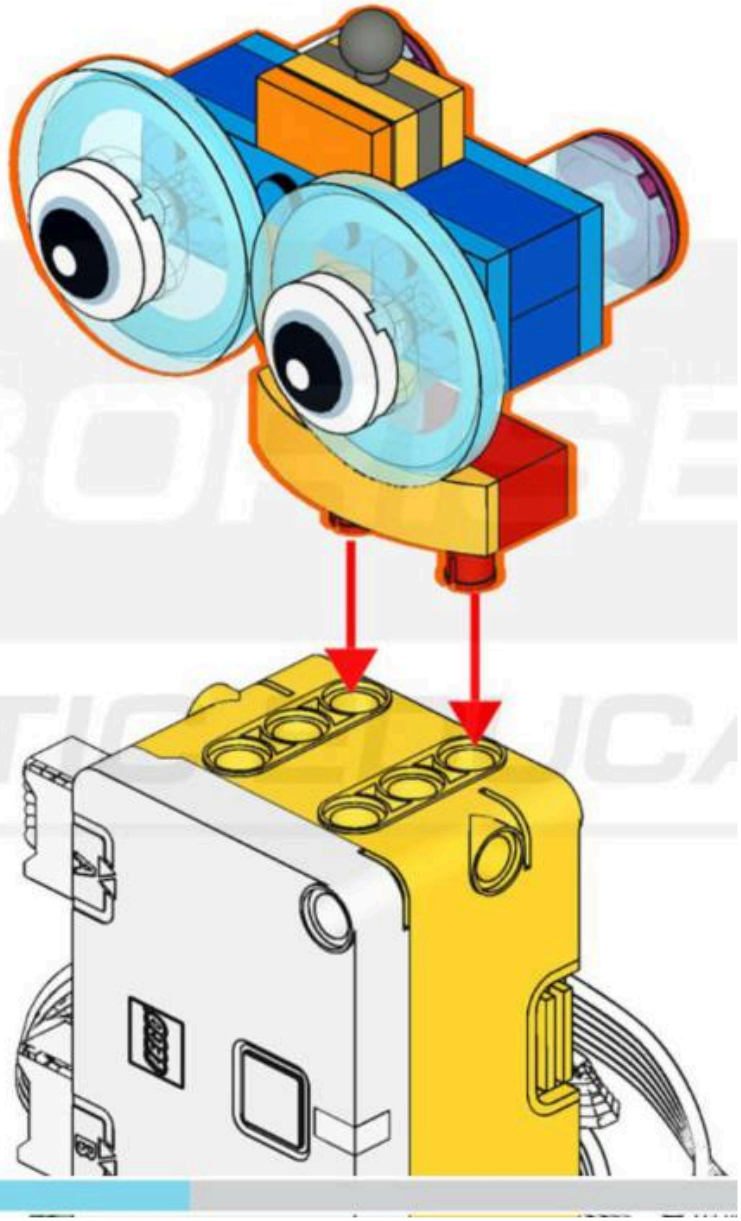


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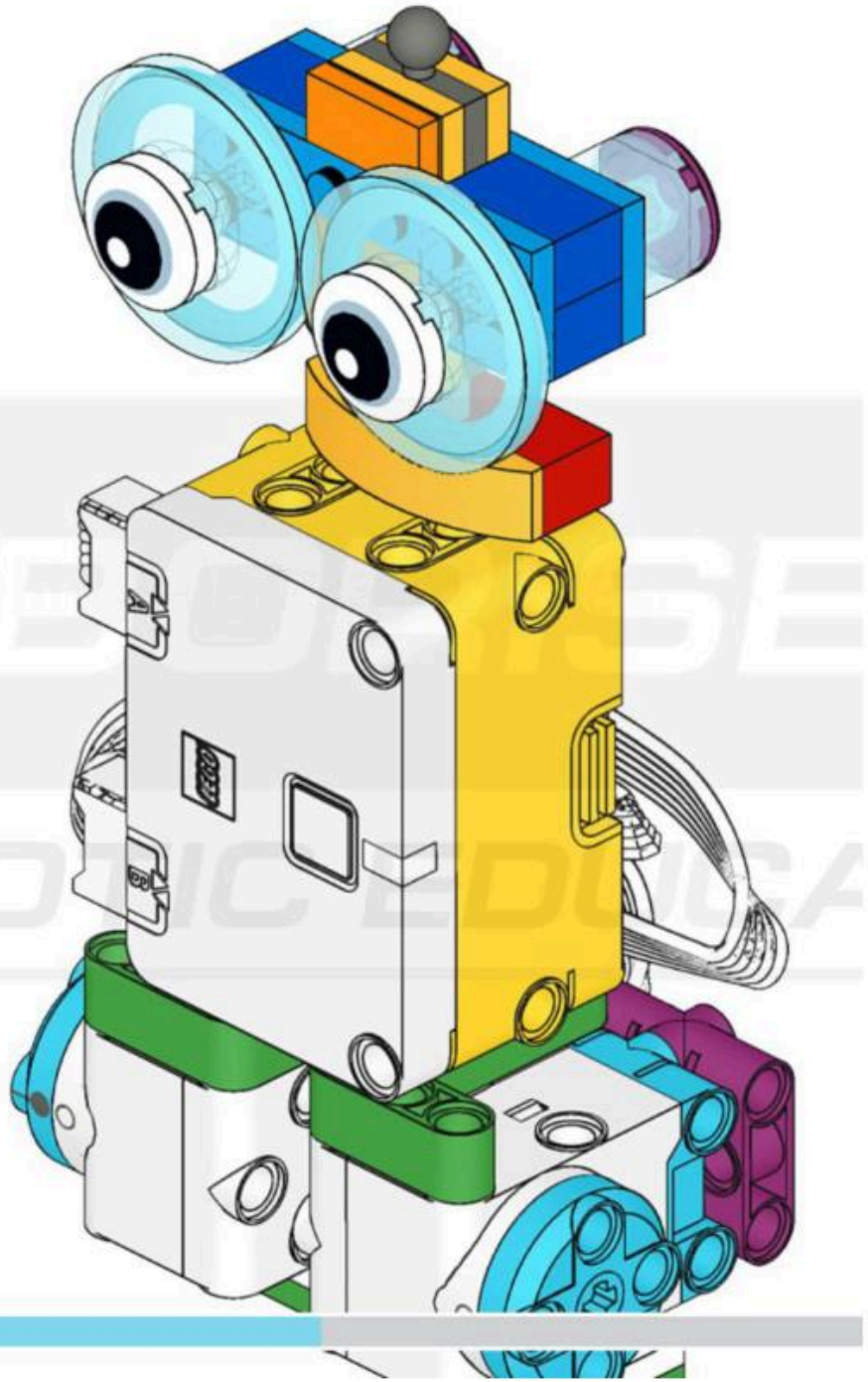
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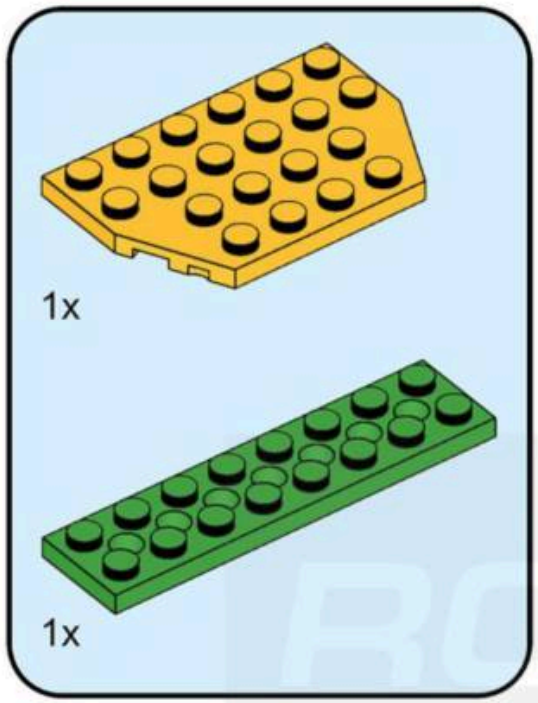


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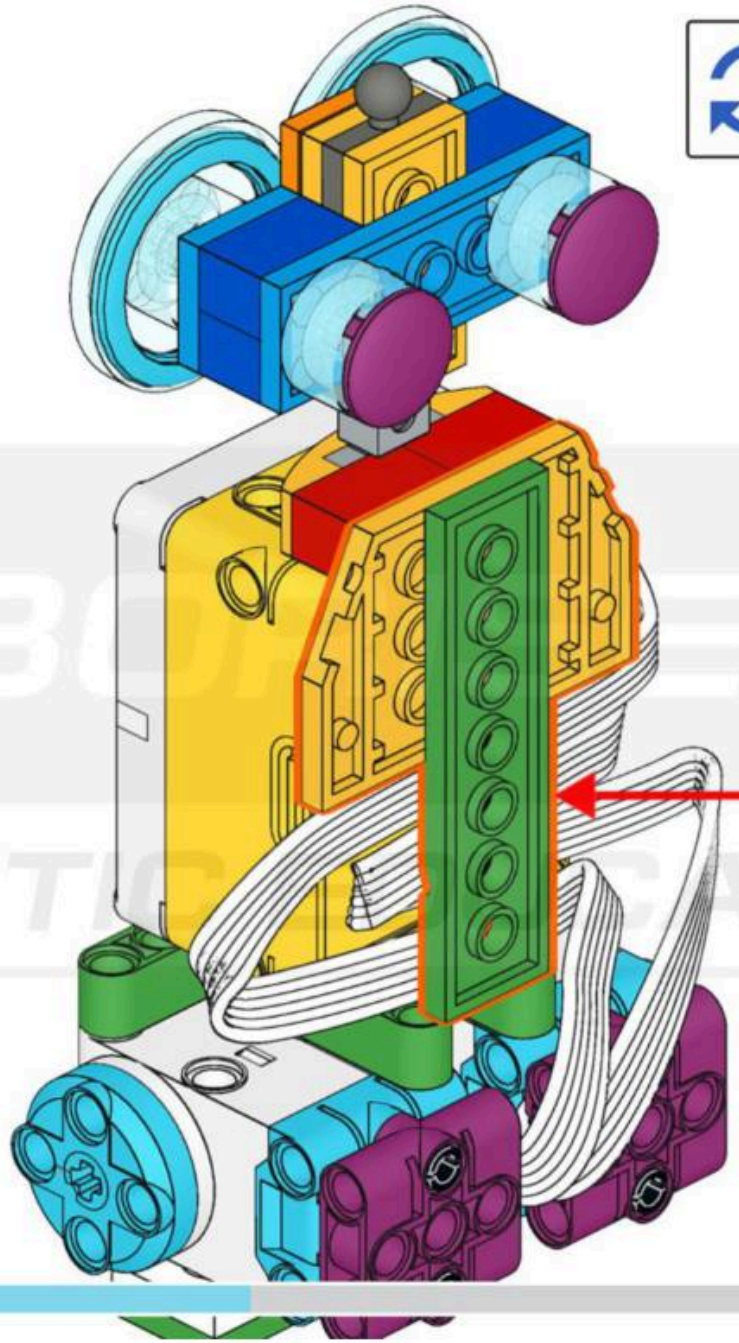


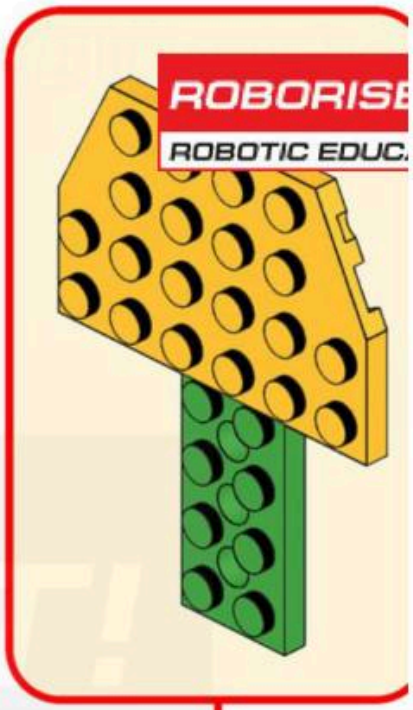
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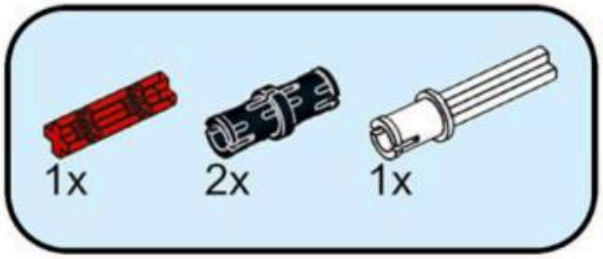




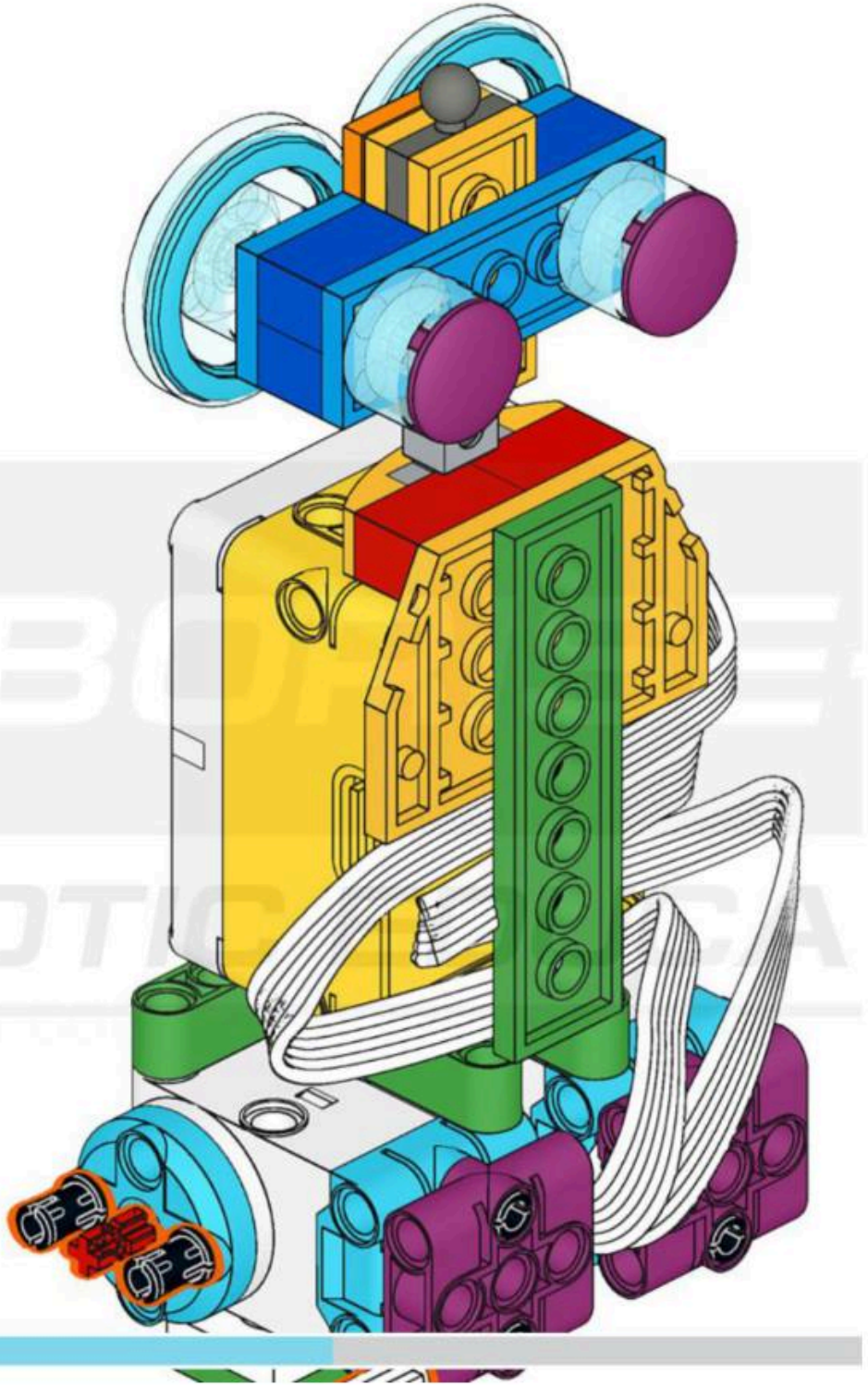
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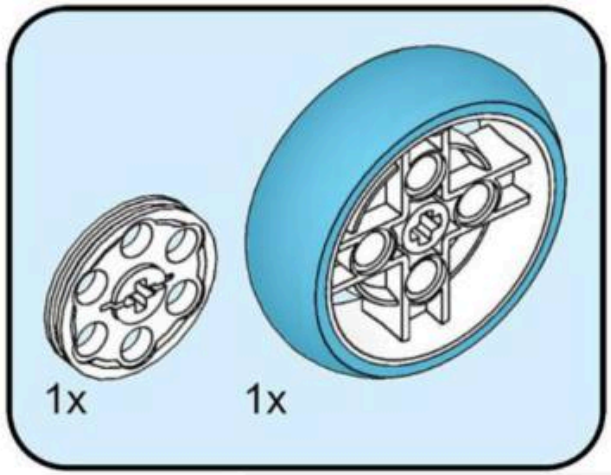




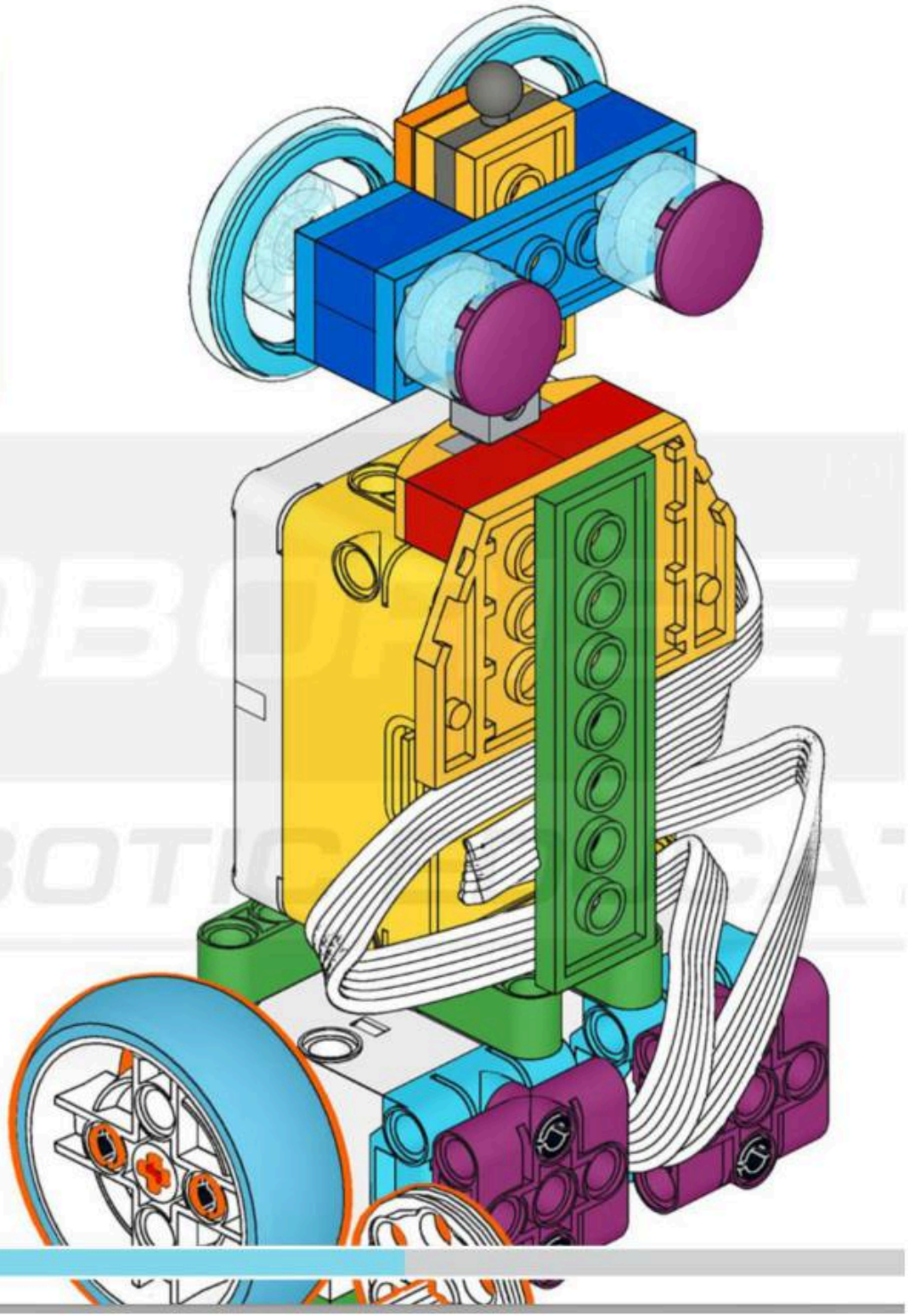


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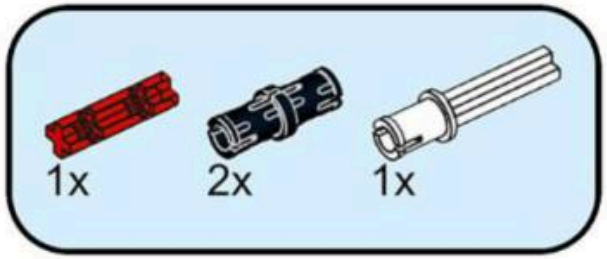




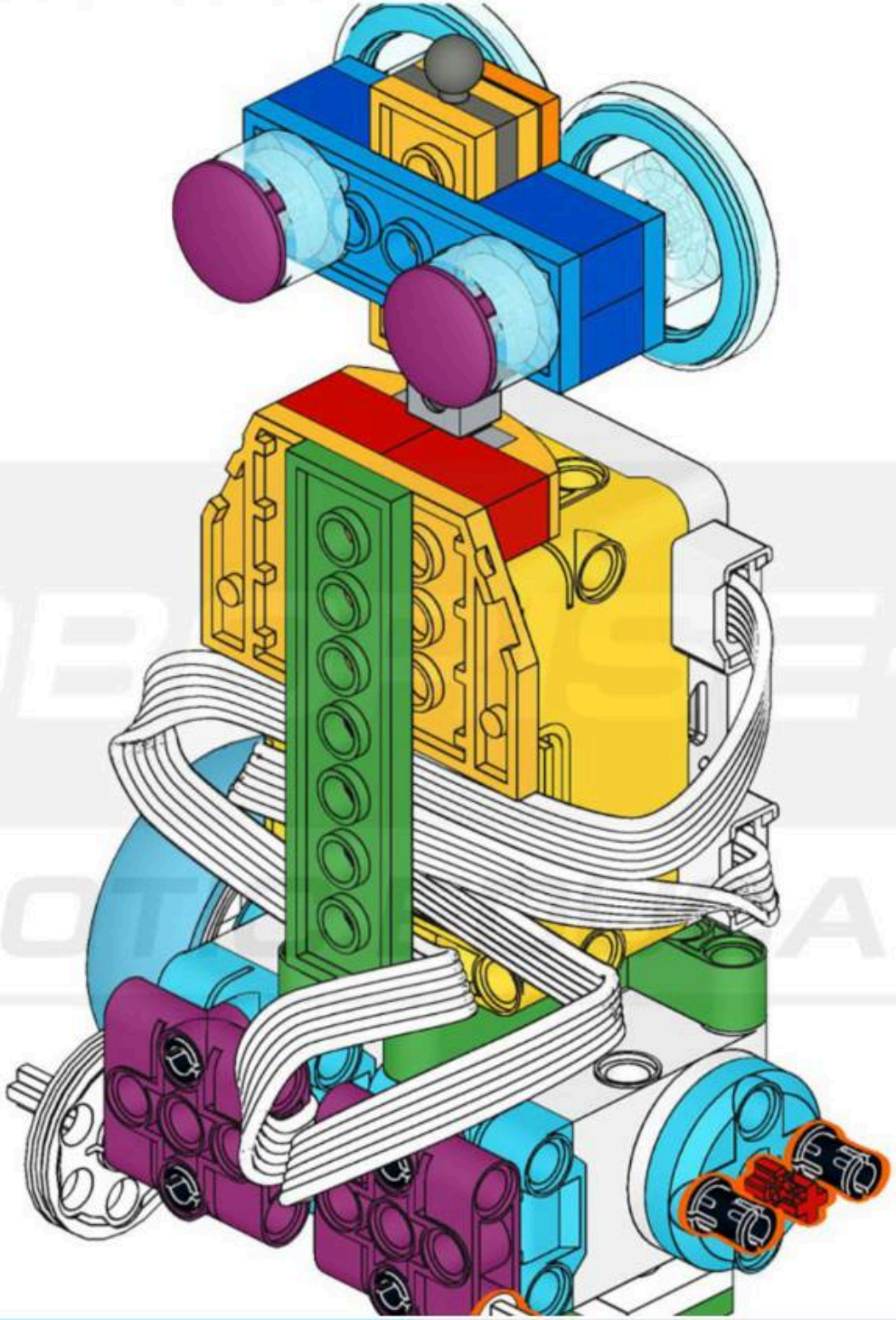
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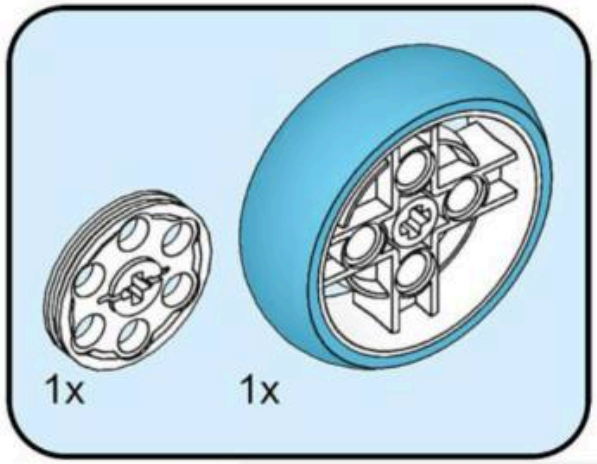


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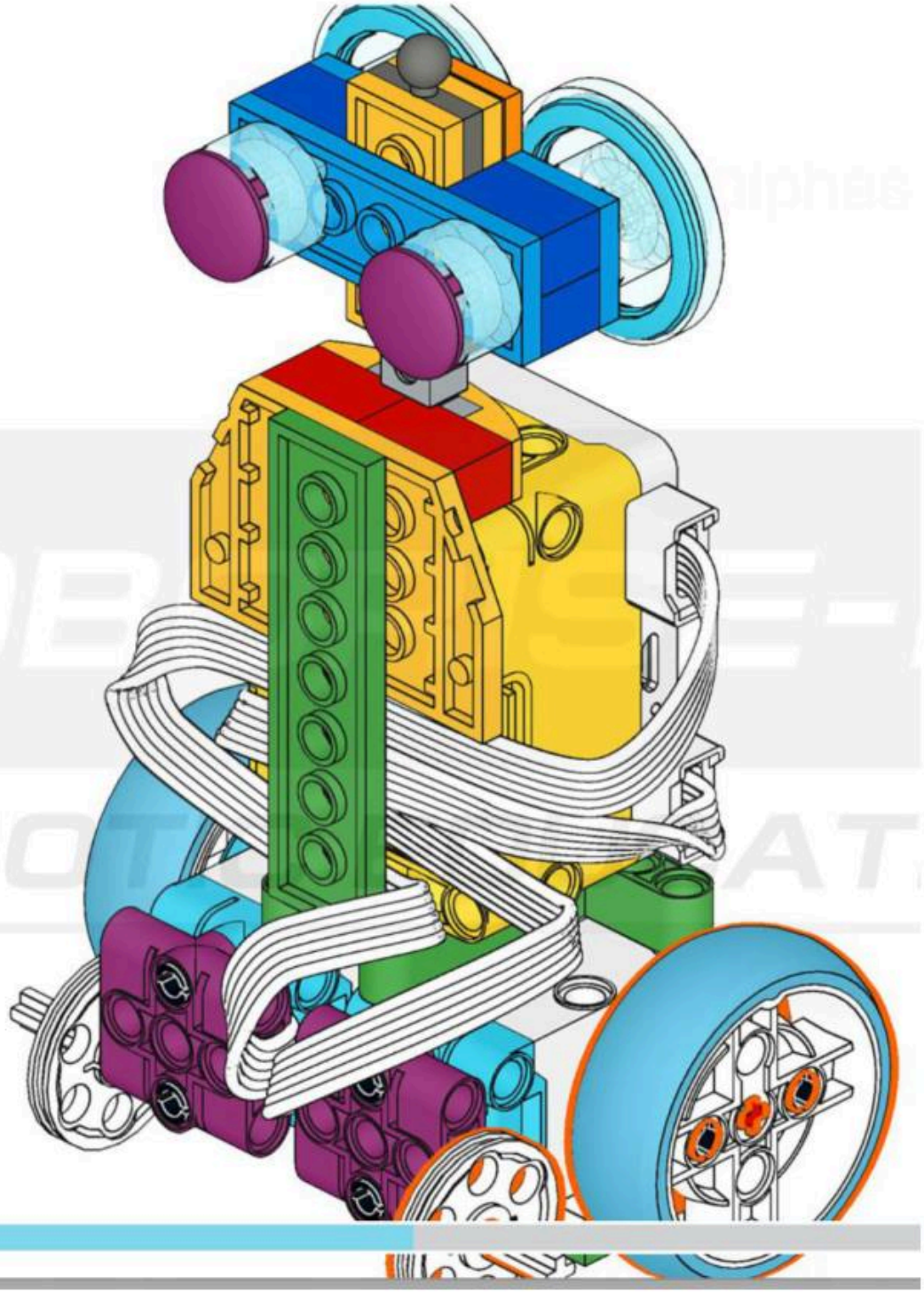


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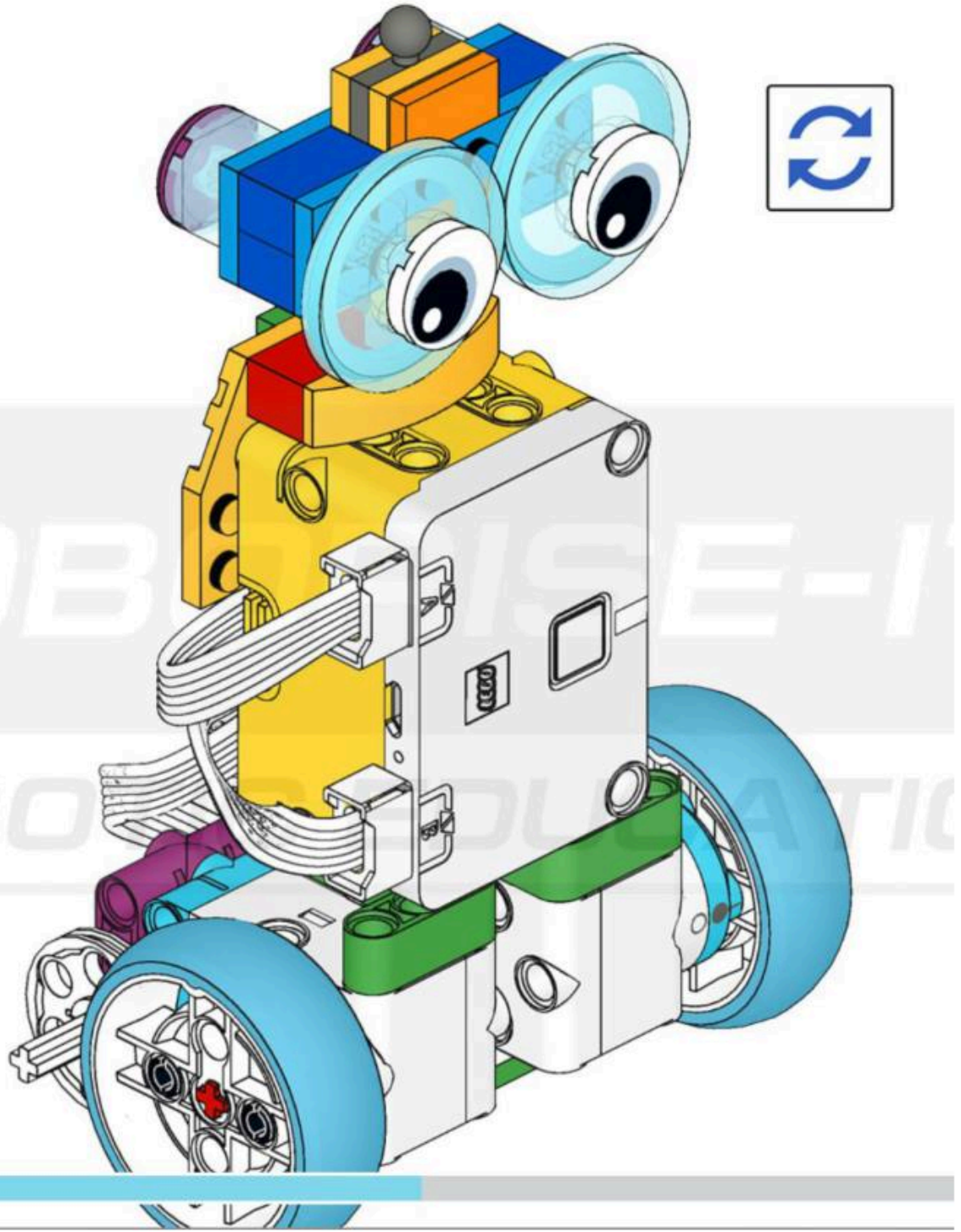




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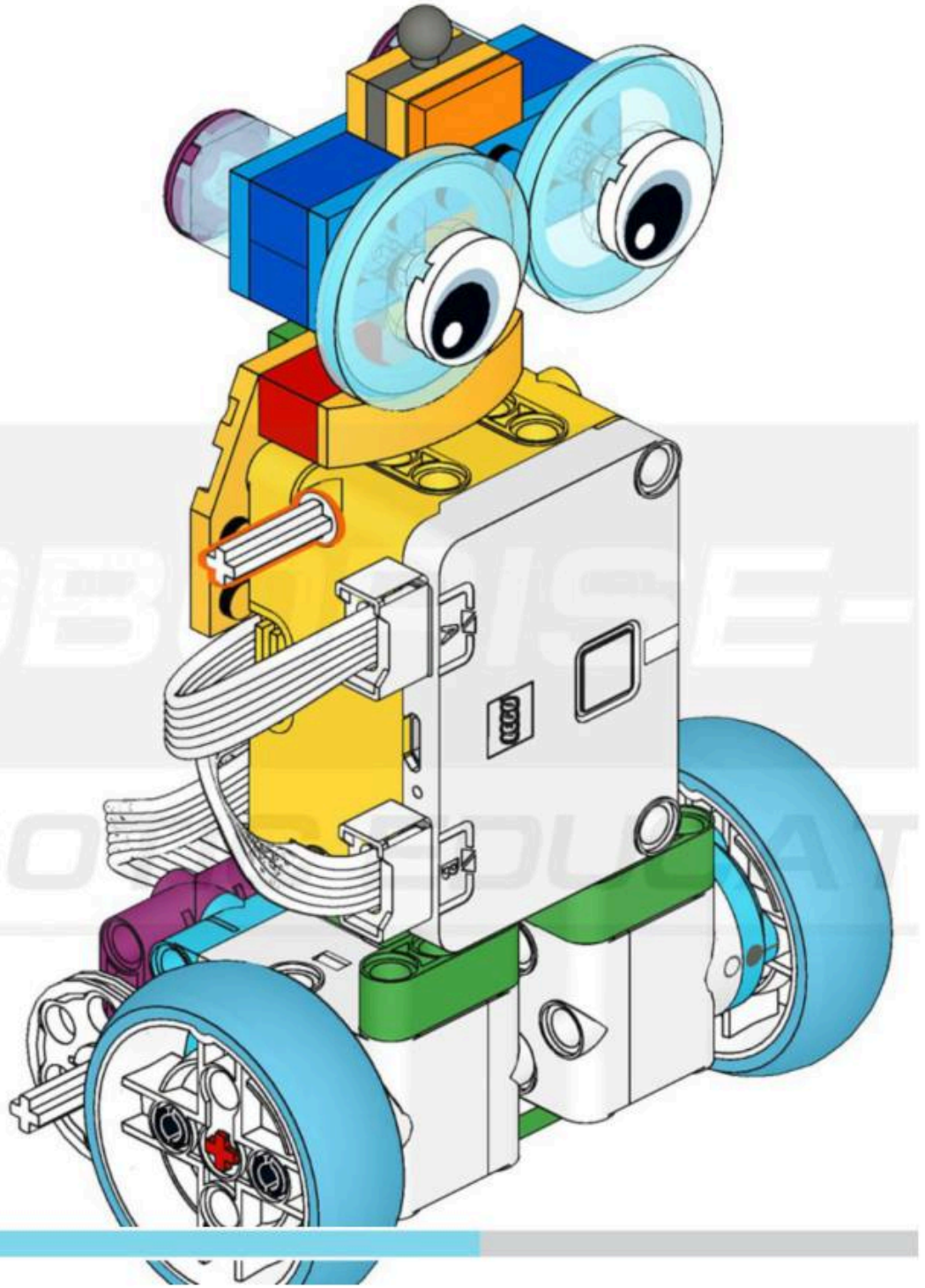


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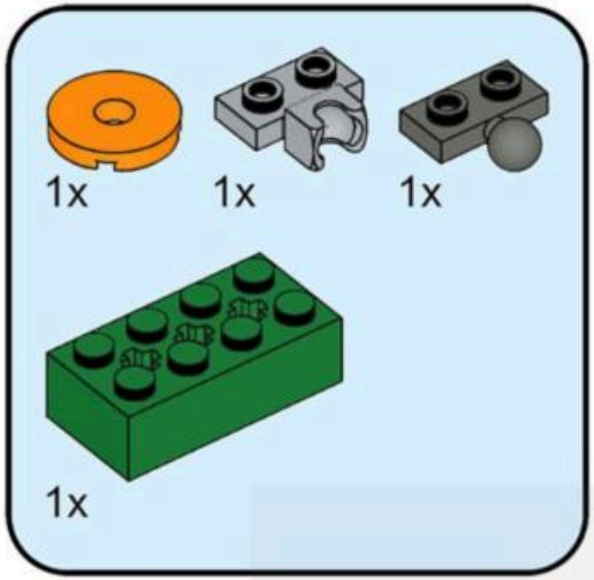




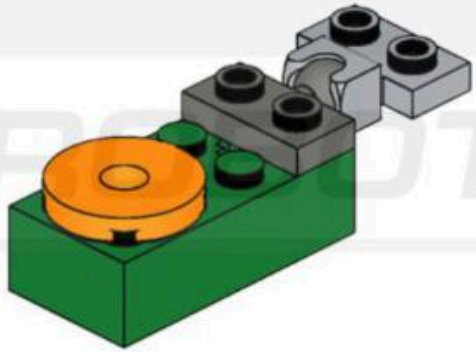
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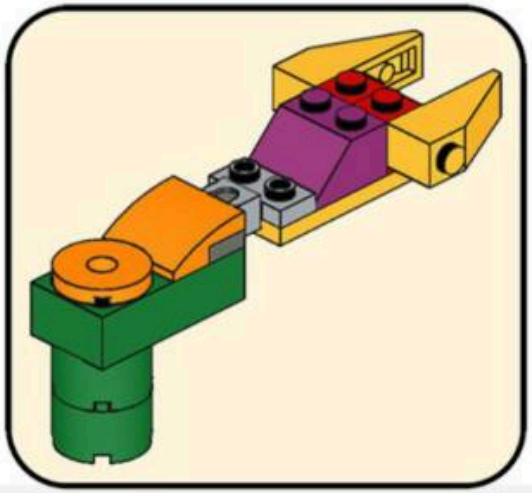


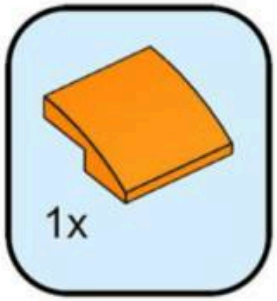
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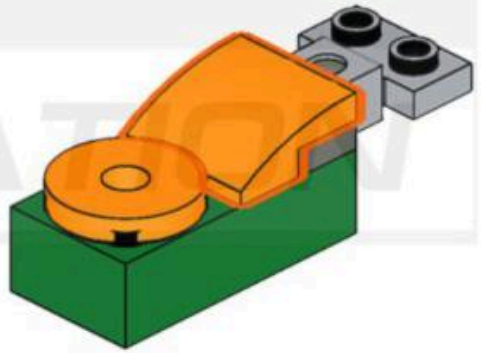
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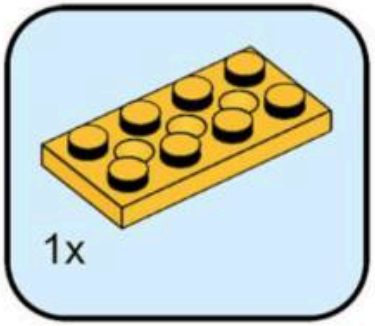




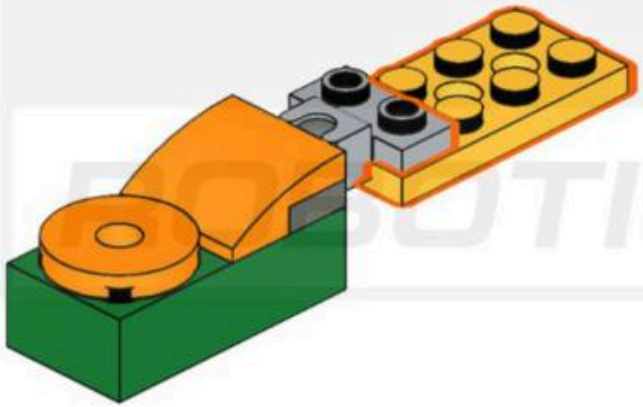
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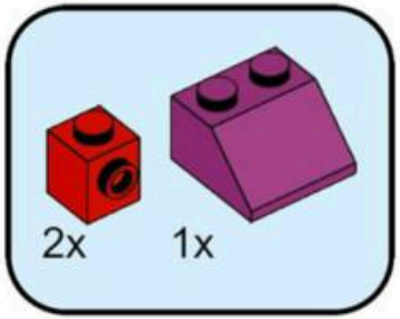


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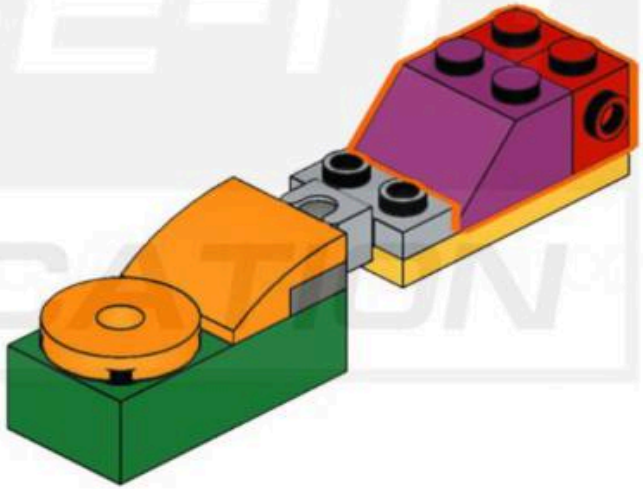


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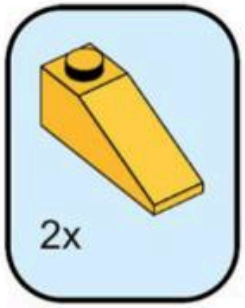




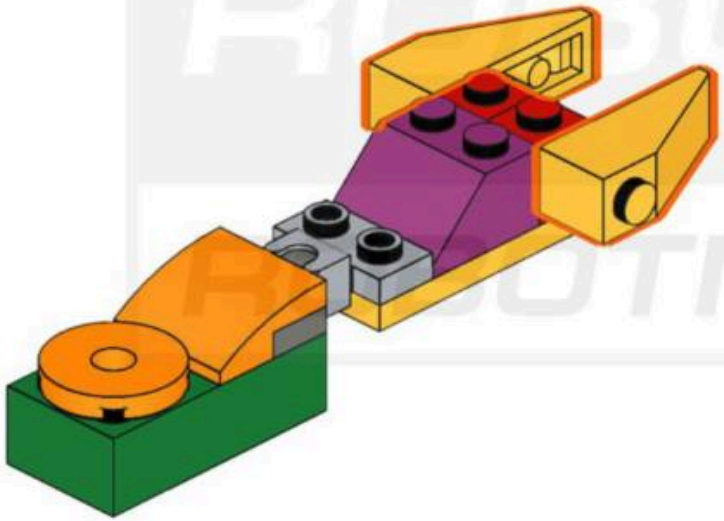
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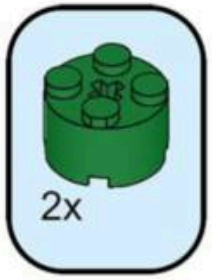


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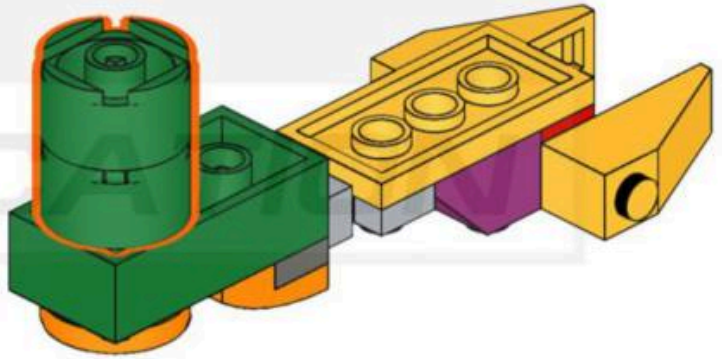


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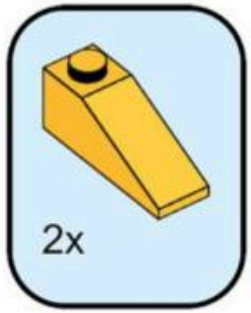




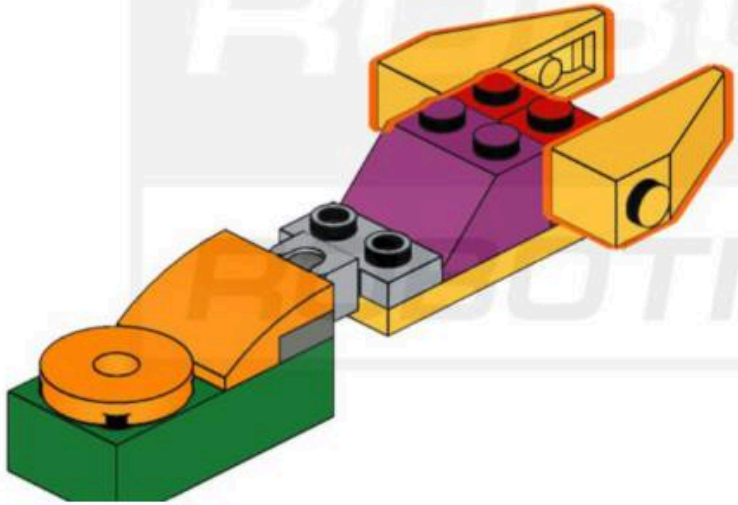
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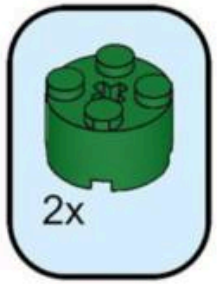


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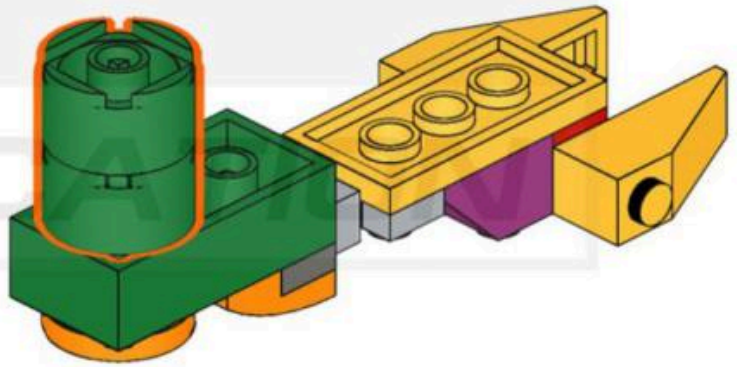


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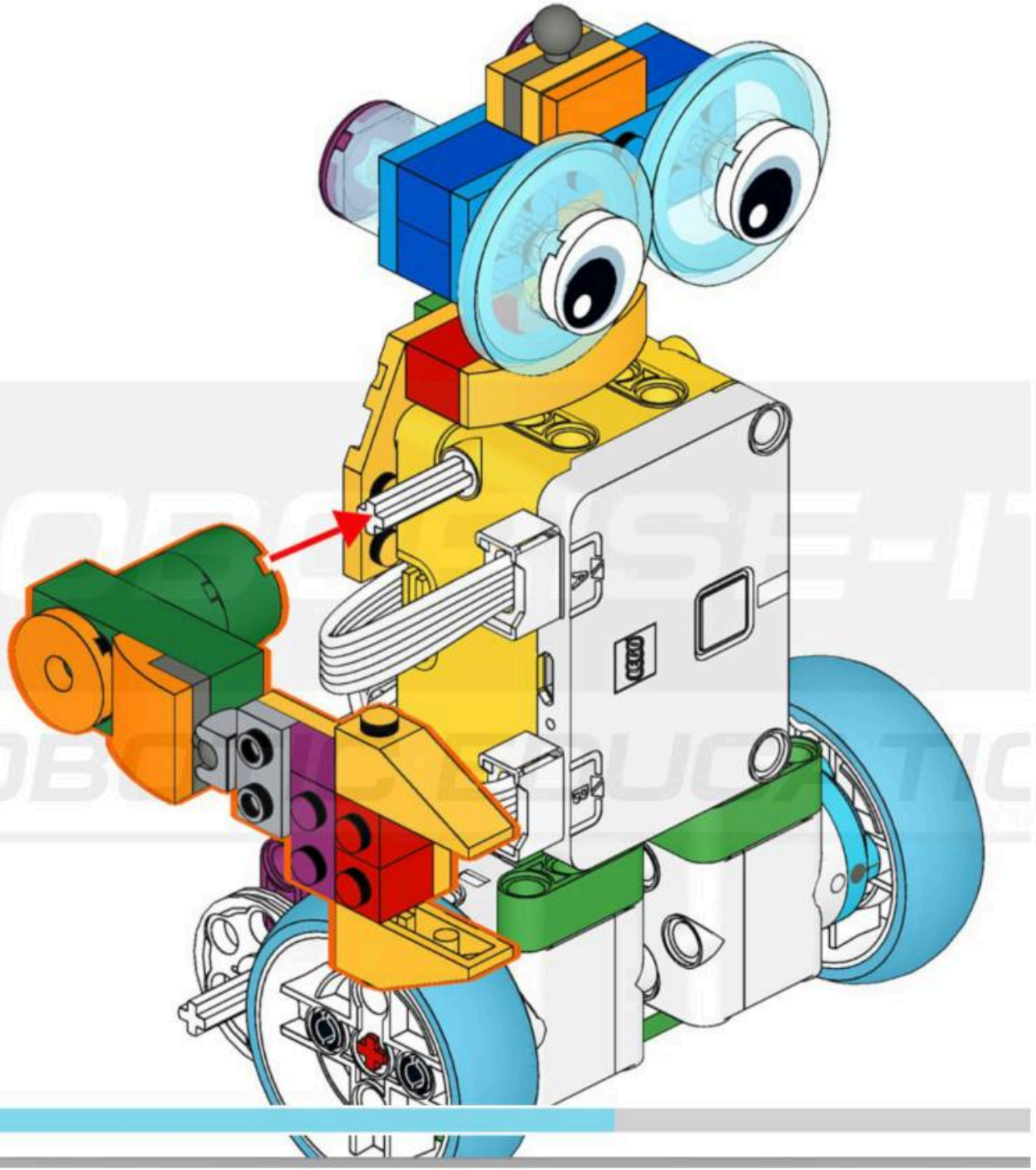


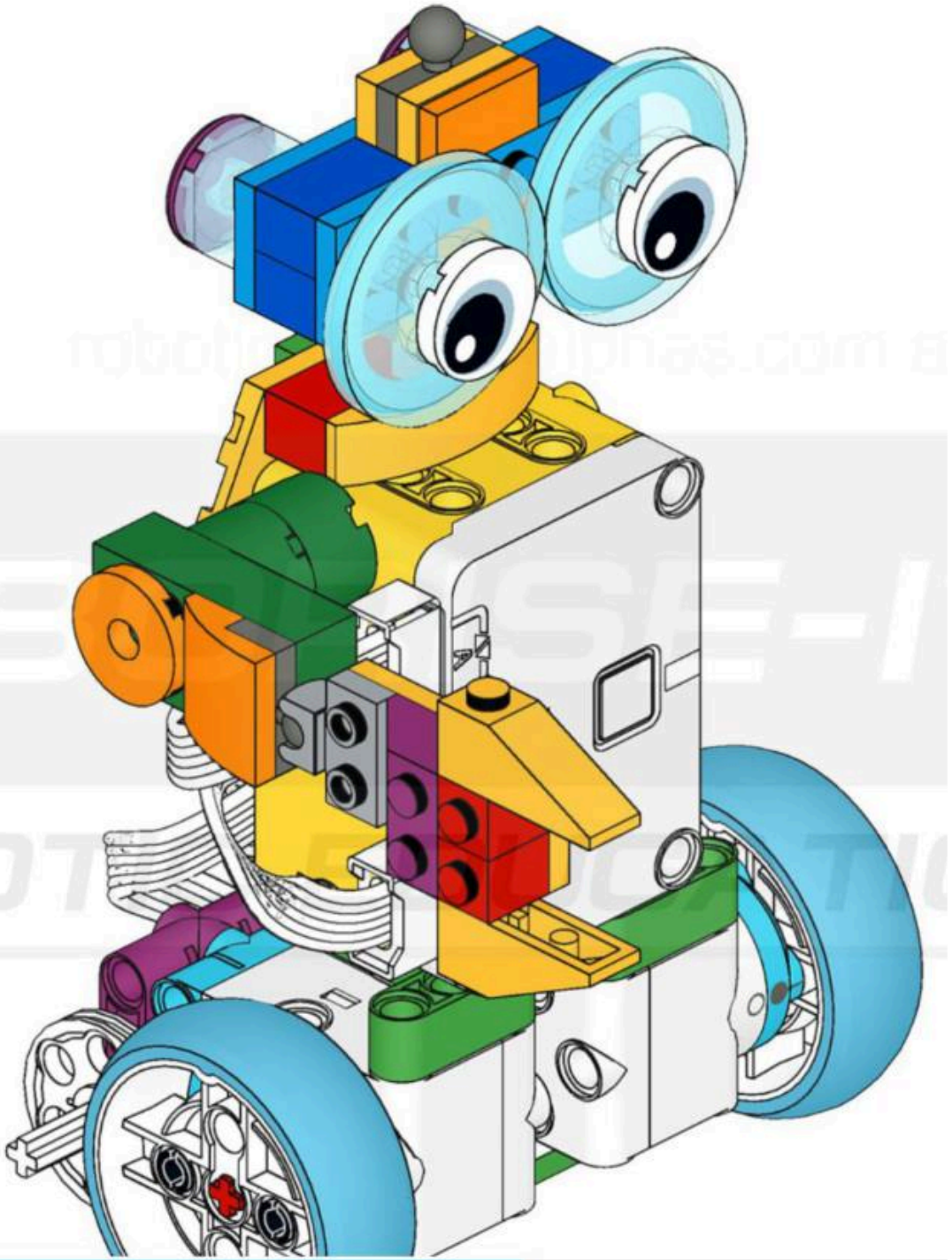
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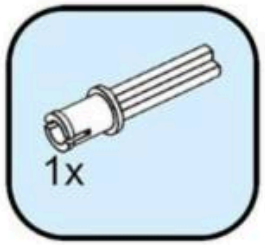


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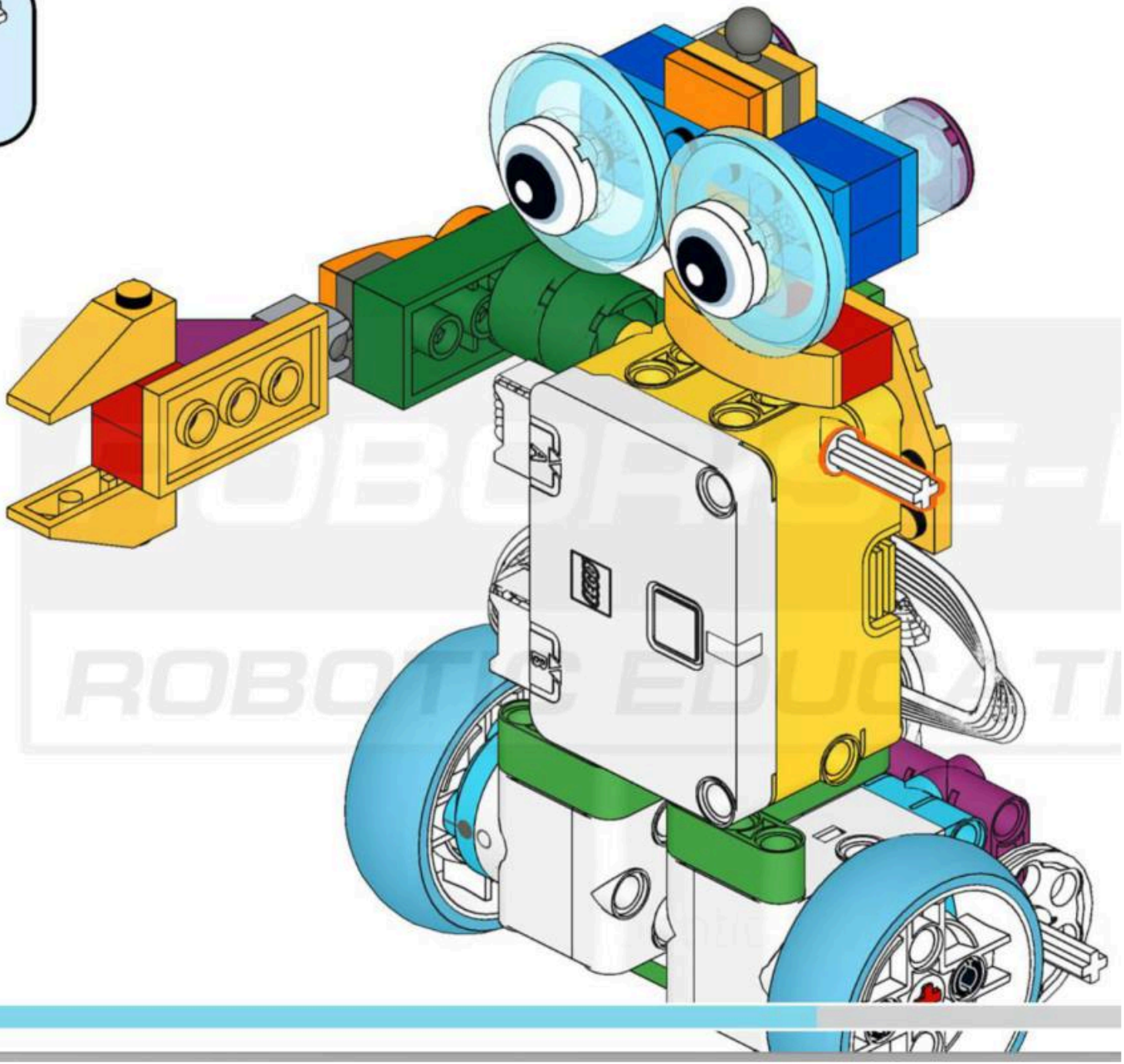
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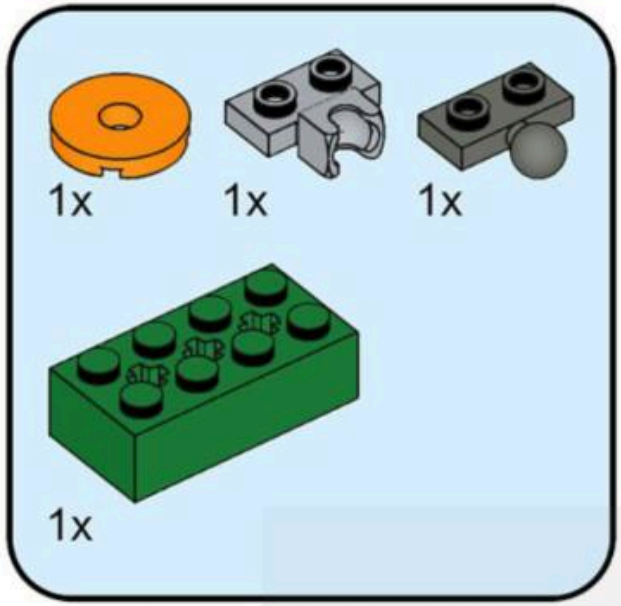




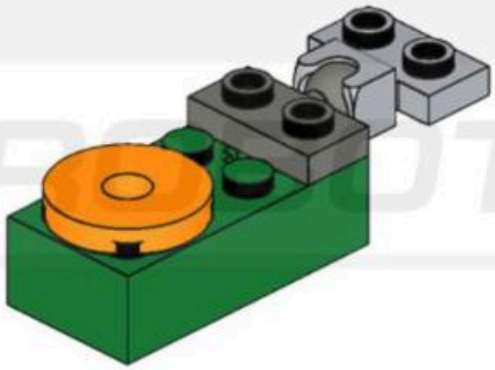
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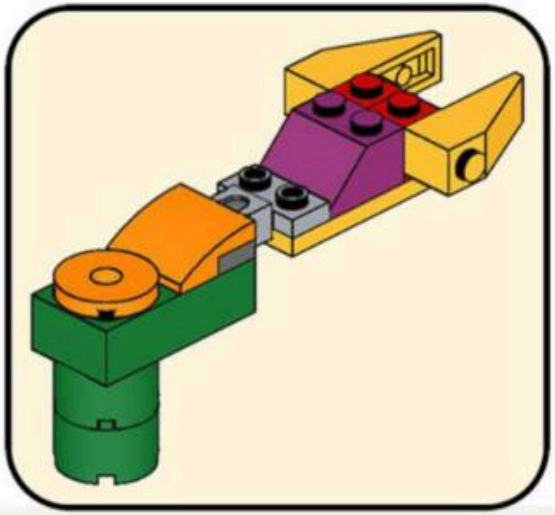


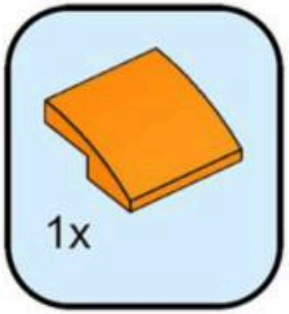
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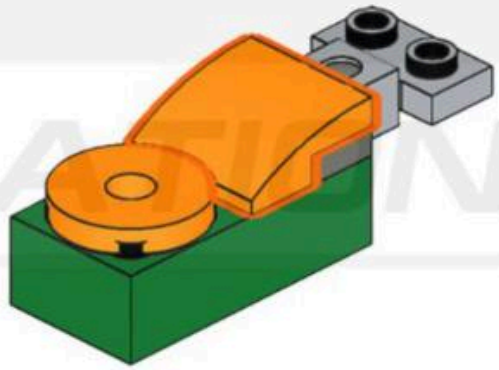
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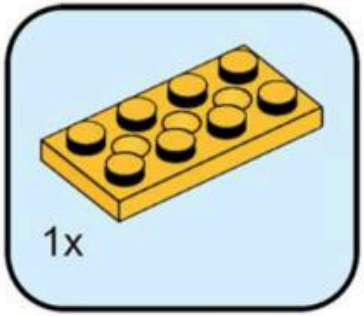




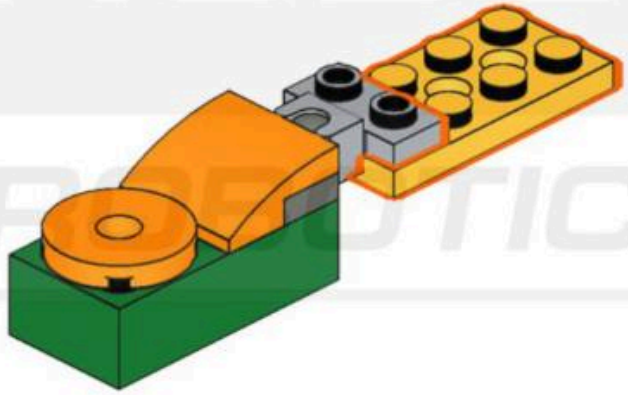
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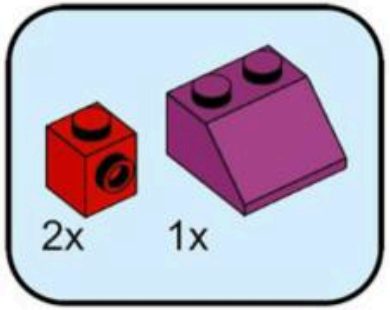


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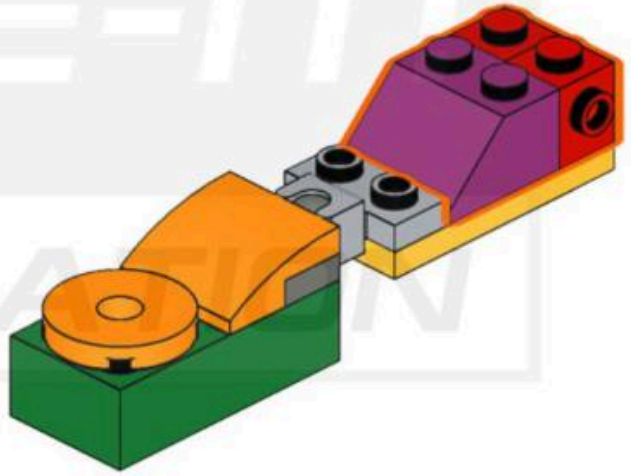


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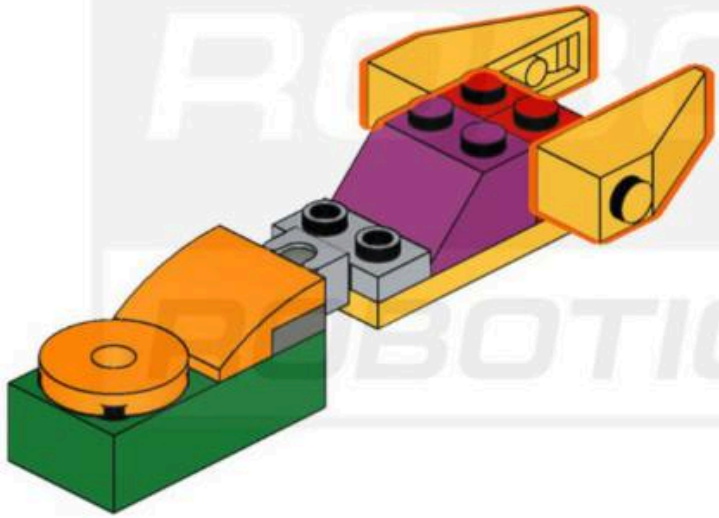
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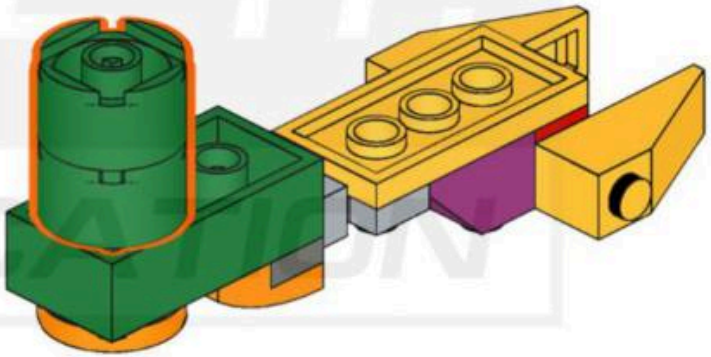
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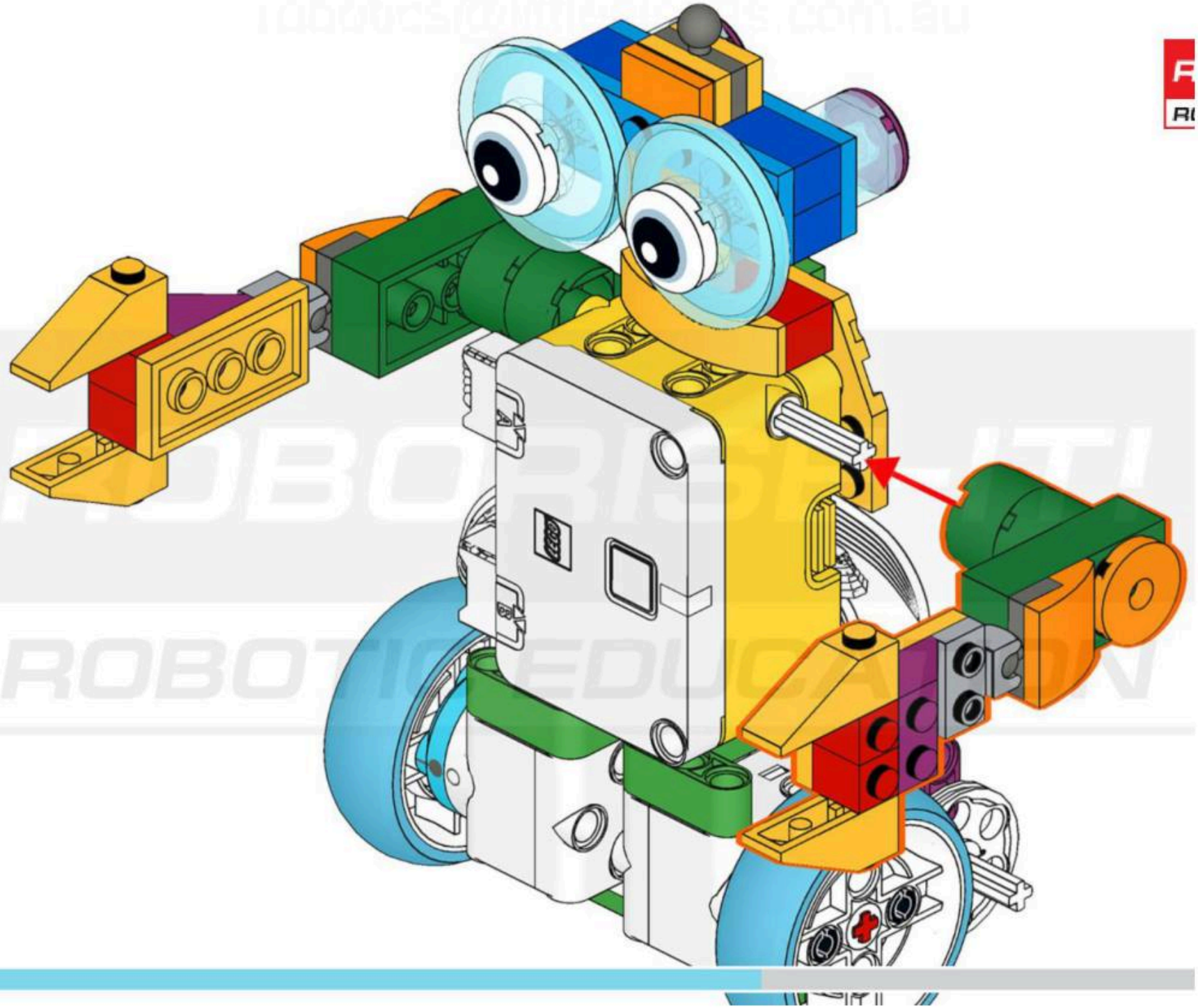
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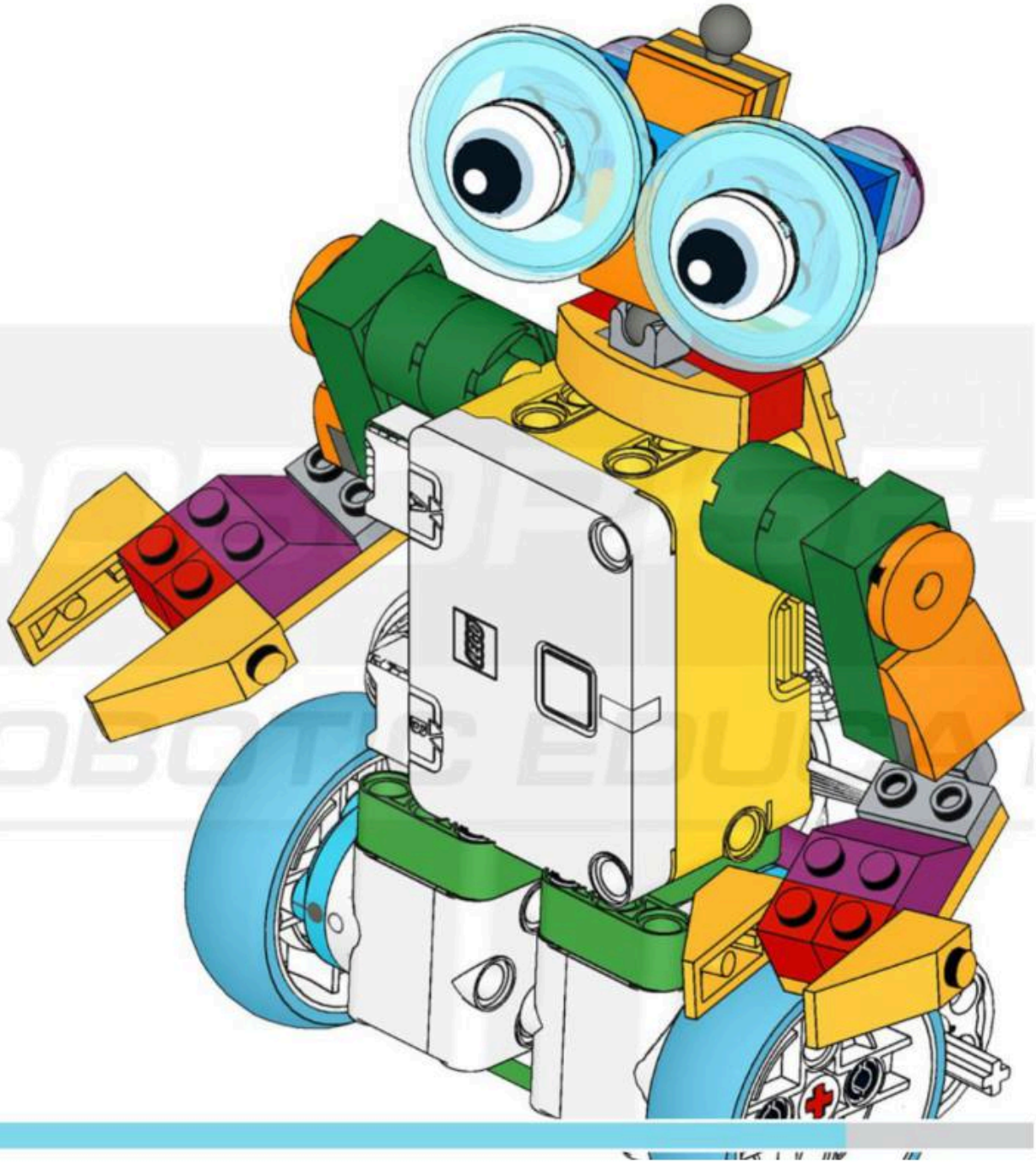


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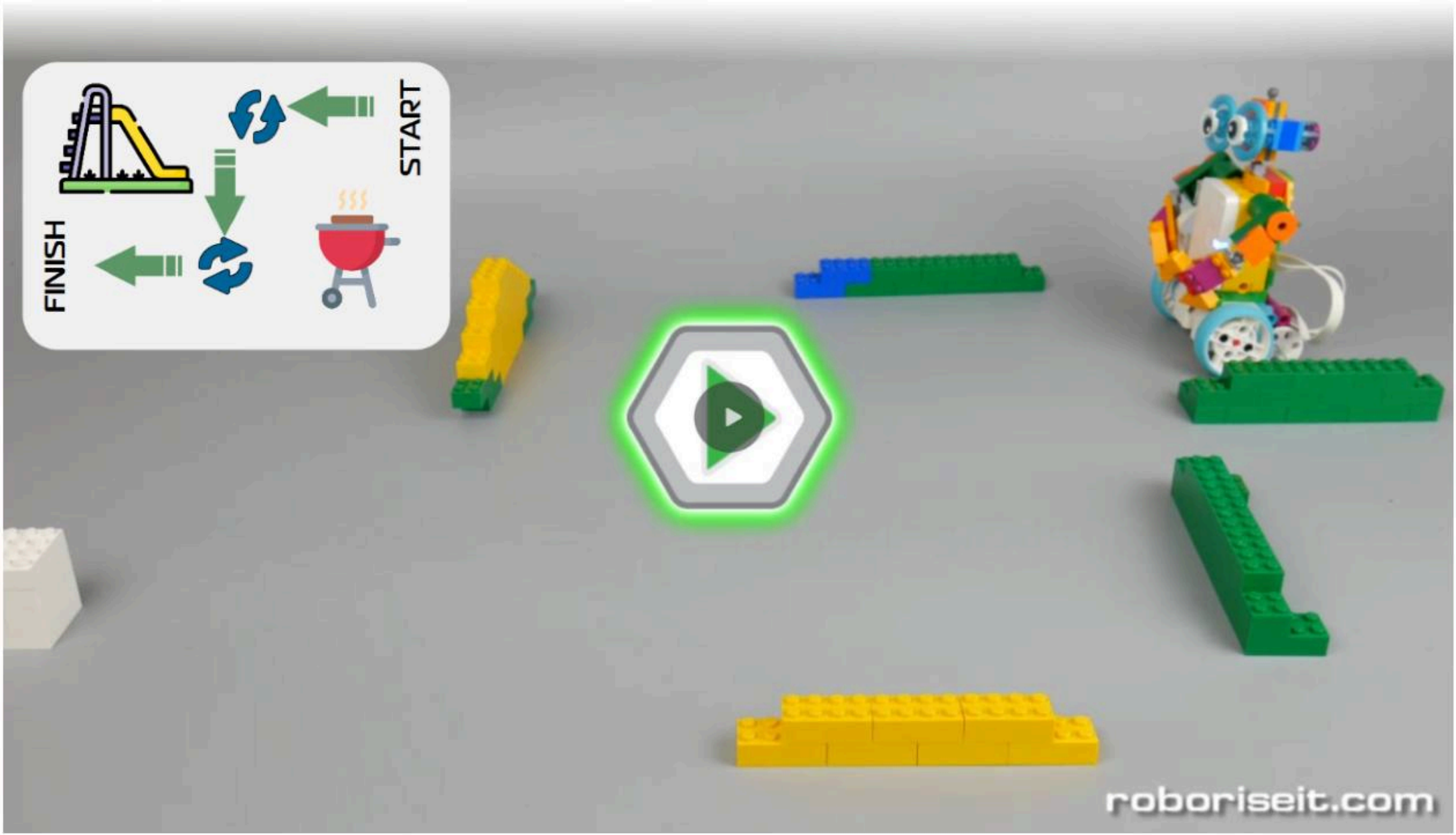
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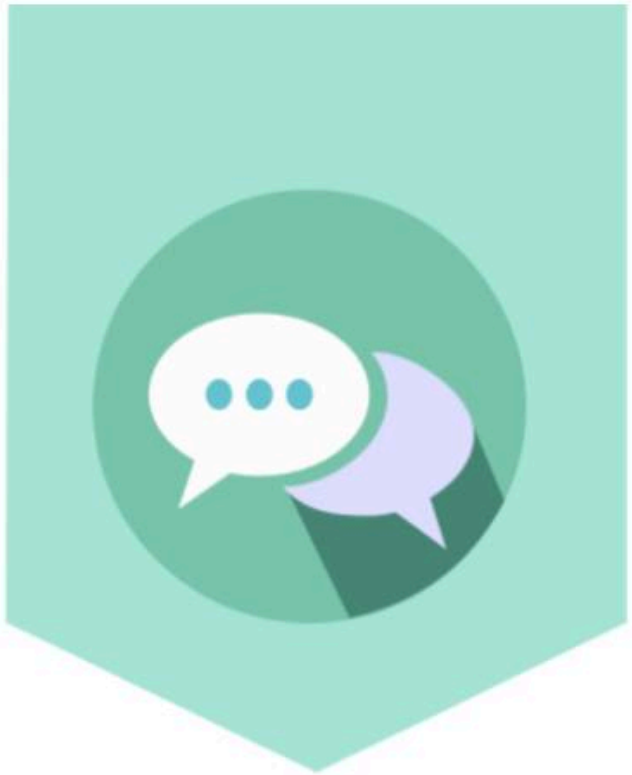


48



Let's program a robot to mow the grass in the yard. Pay attention to the location of the barbecue area. It is desirable that the robot does not destroy it!



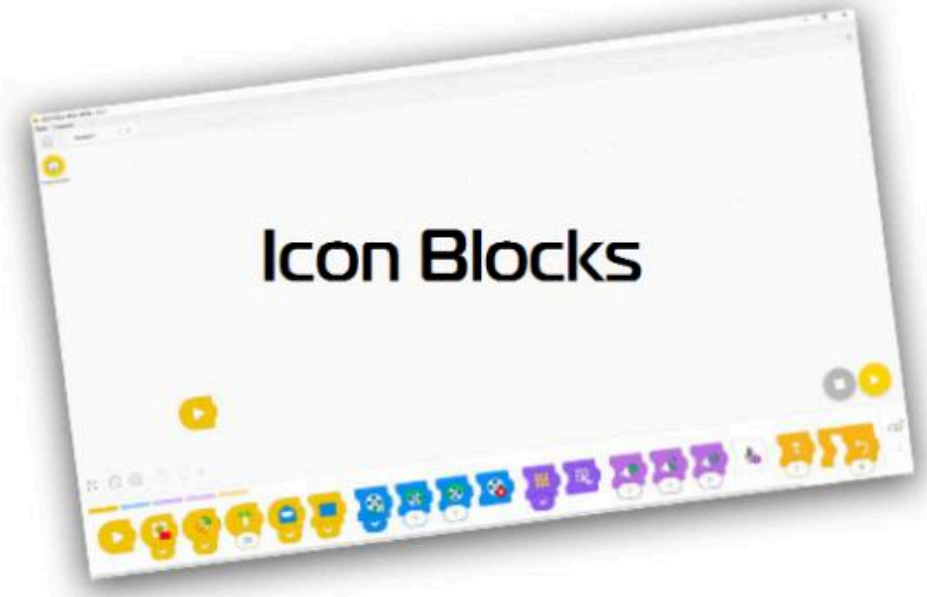


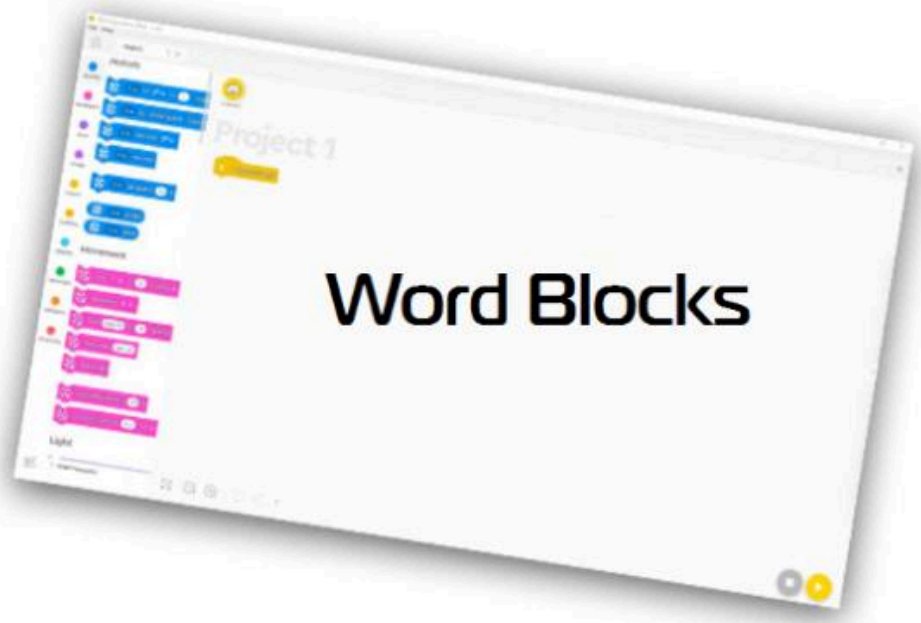
What movements should a robot be able to perform in order to perform such a task?



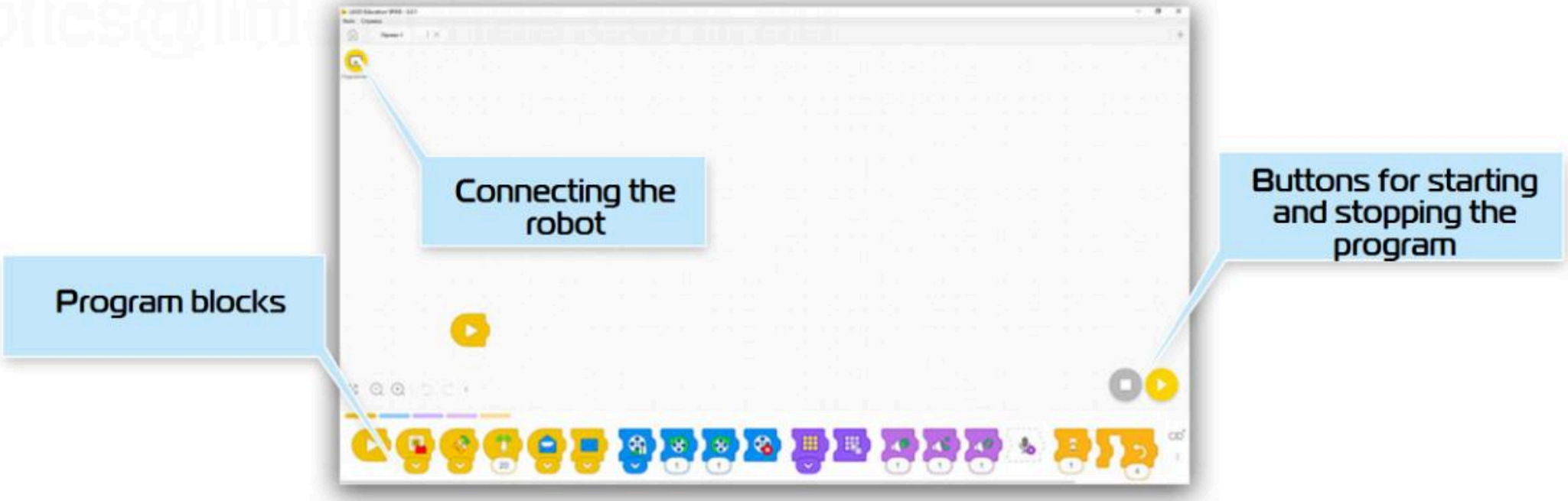
In order to make programming easy and convenient, special programs - programming environments - are created. Thanks to them, the programmer's communication and work takes place in a language that is understandable to both.

Let's get to know the SPIKE Essential robot programming environment. You can create apps in either Icon Blocks mode or Word Blocks mode.



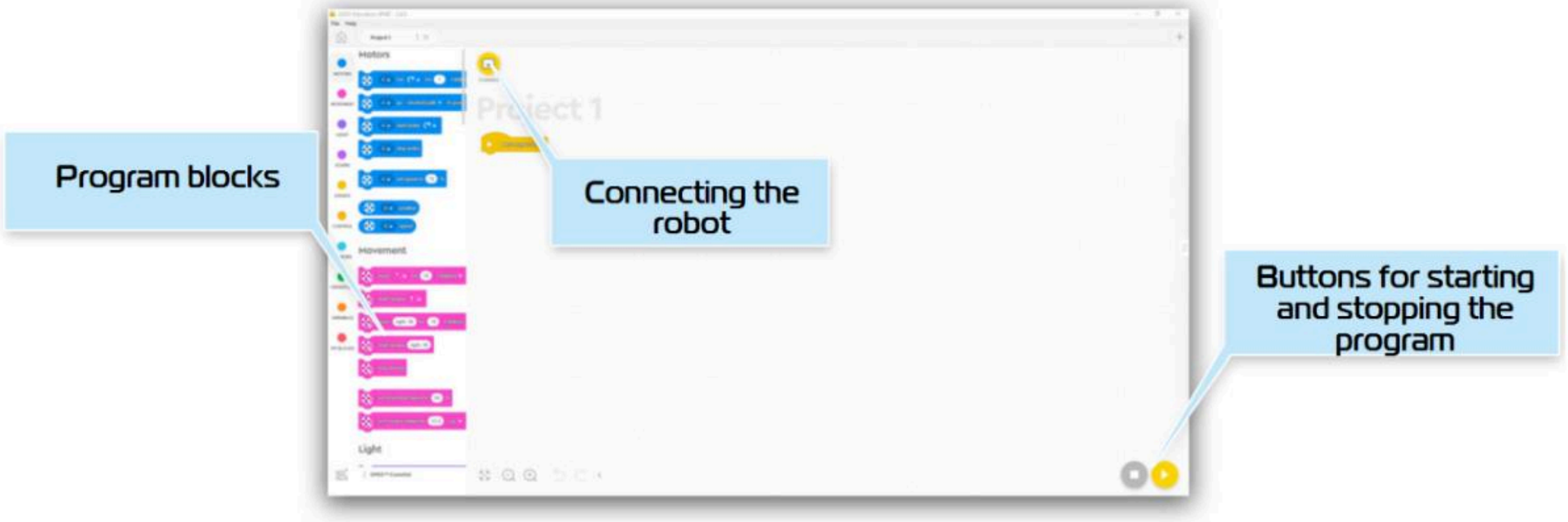


Icon Blocks mode is simpler, but has very limited functionality. That is, not everything that we would like to teach our robots can be programmed in this mode.



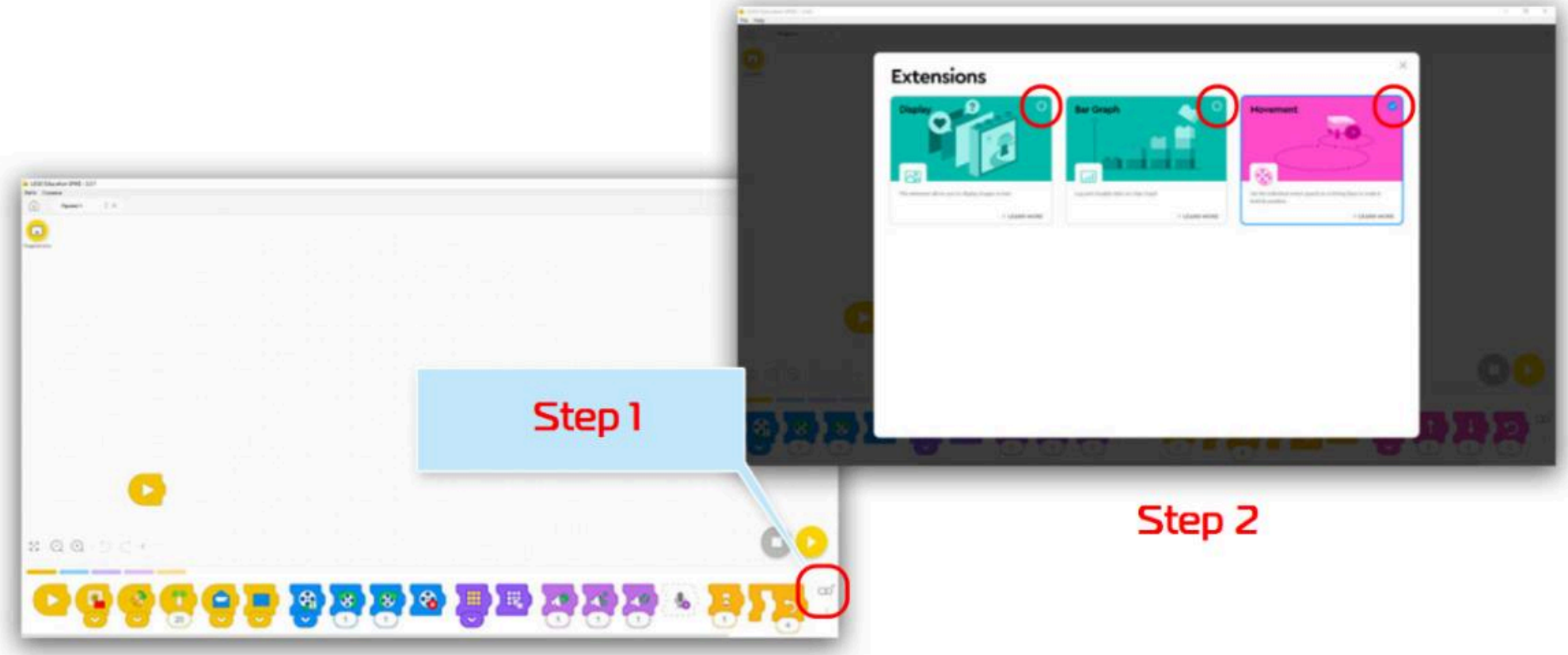
The program consists of program blocks that are executed from left to right

Instead, World Blocks mode allows you to program much more complex and interesting robot behavior. There are more software blocks and opportunities in general.

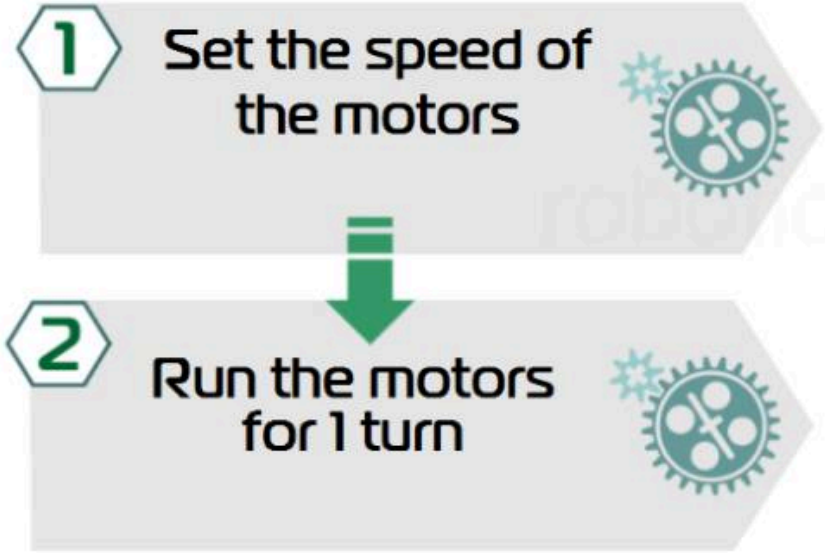


Here, the program blocks are executed from top to bottom

True robot programmers have always been limited by standard programs, so even more blocks can be added to the programming environment! Here's how it's done in ICON Blocks mode:



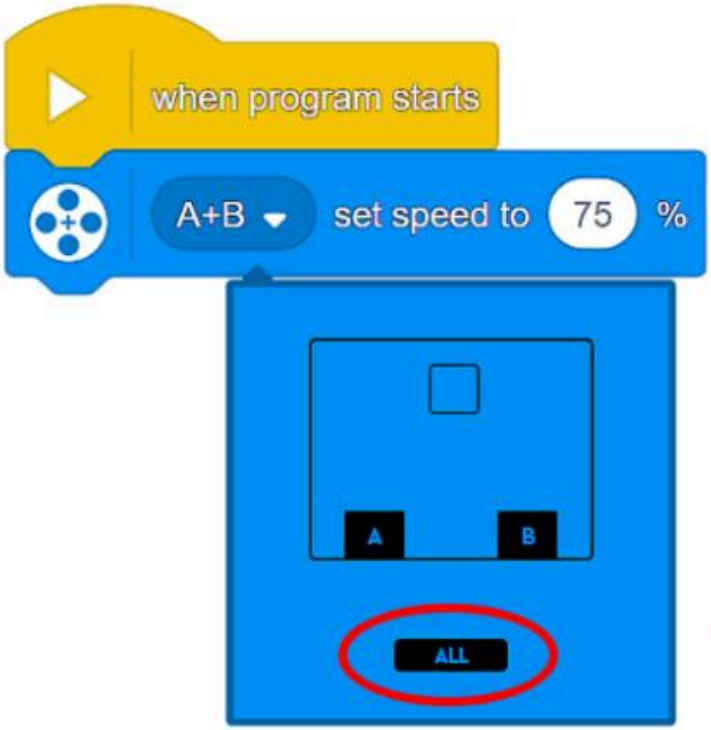
The program should work according to the following algorithm:

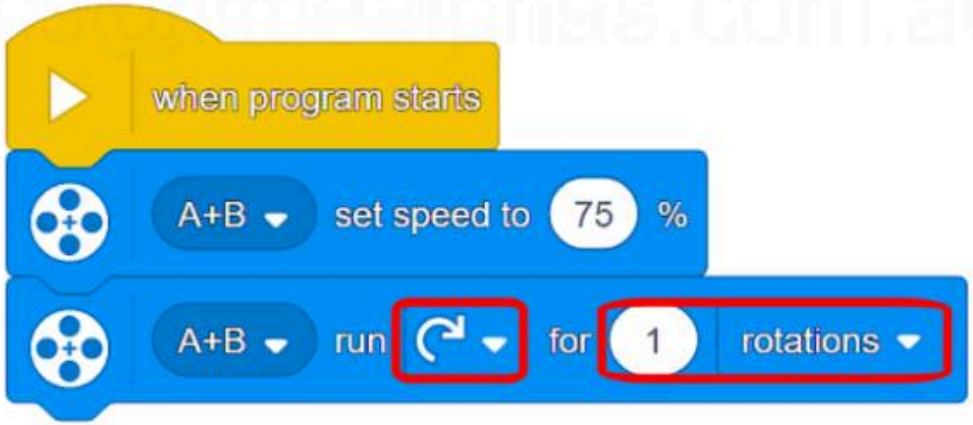


An algorithm is a sequence of actions that leads to the completion of a task. The algorithm is turned into a program, understandable work, and the program is already executed by the robot

Use blocks to start motors:

In Word Blocks mode, the program looks like this:





In the second block, set the rotation direction and duration

To enable both motors, select the 'All' option

Use blocks to start motors:

In ICON Blocks mode, the program looks like this:



During testing, you could notice that instead of moving forward, the robot began to turn. This happened due to the fact that the software blocks for turning on the motors turn them on in the same direction. And in the model, the motors are mounted in a mirror.



Use the blocks to program the movement of the robot:

In ICON Blocks mode, the program looks like this:



Repeat the robot test

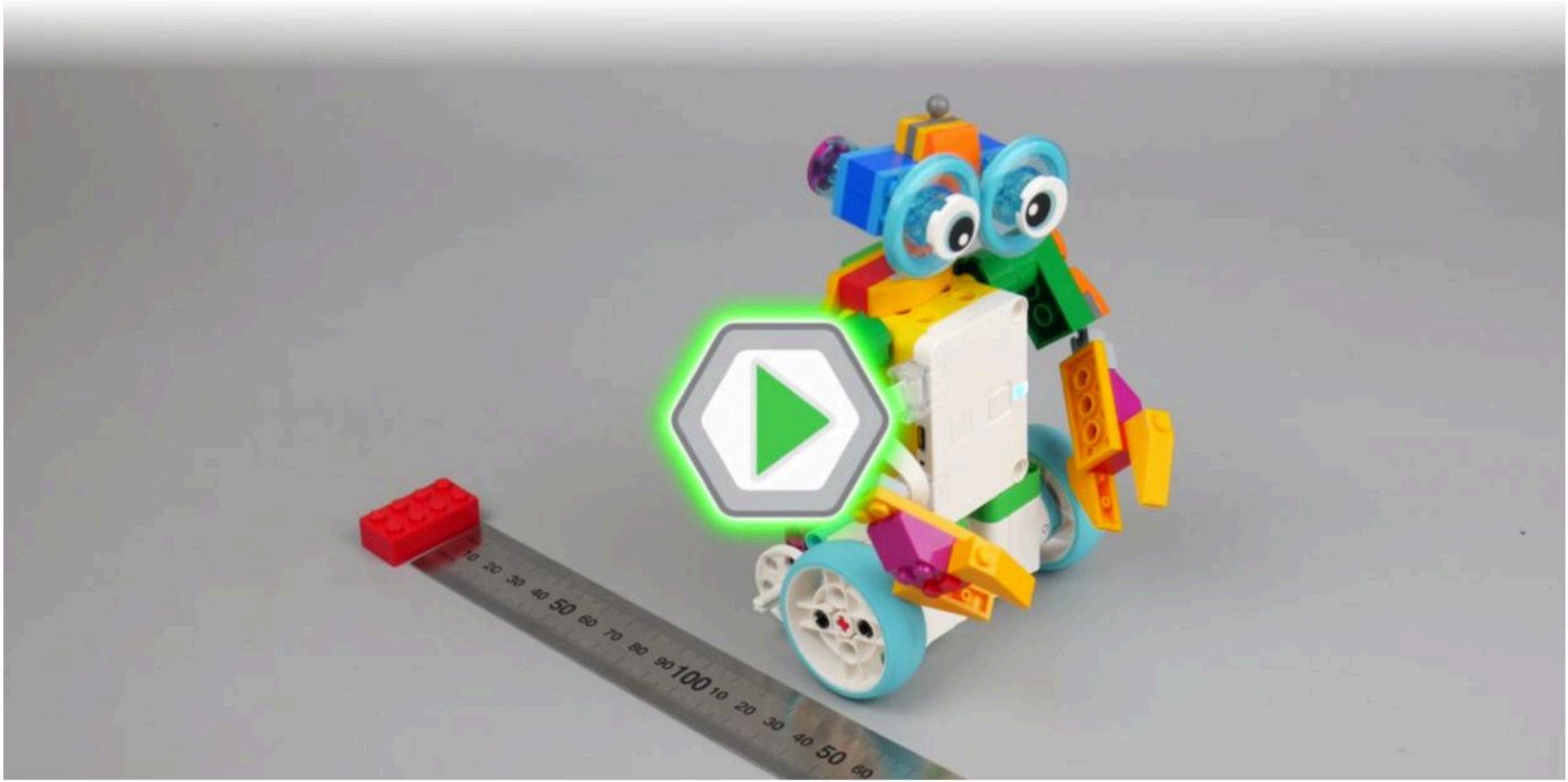
Use the blocks to program the movement of the robot:

In Word Blocks mode, the program looks like this:

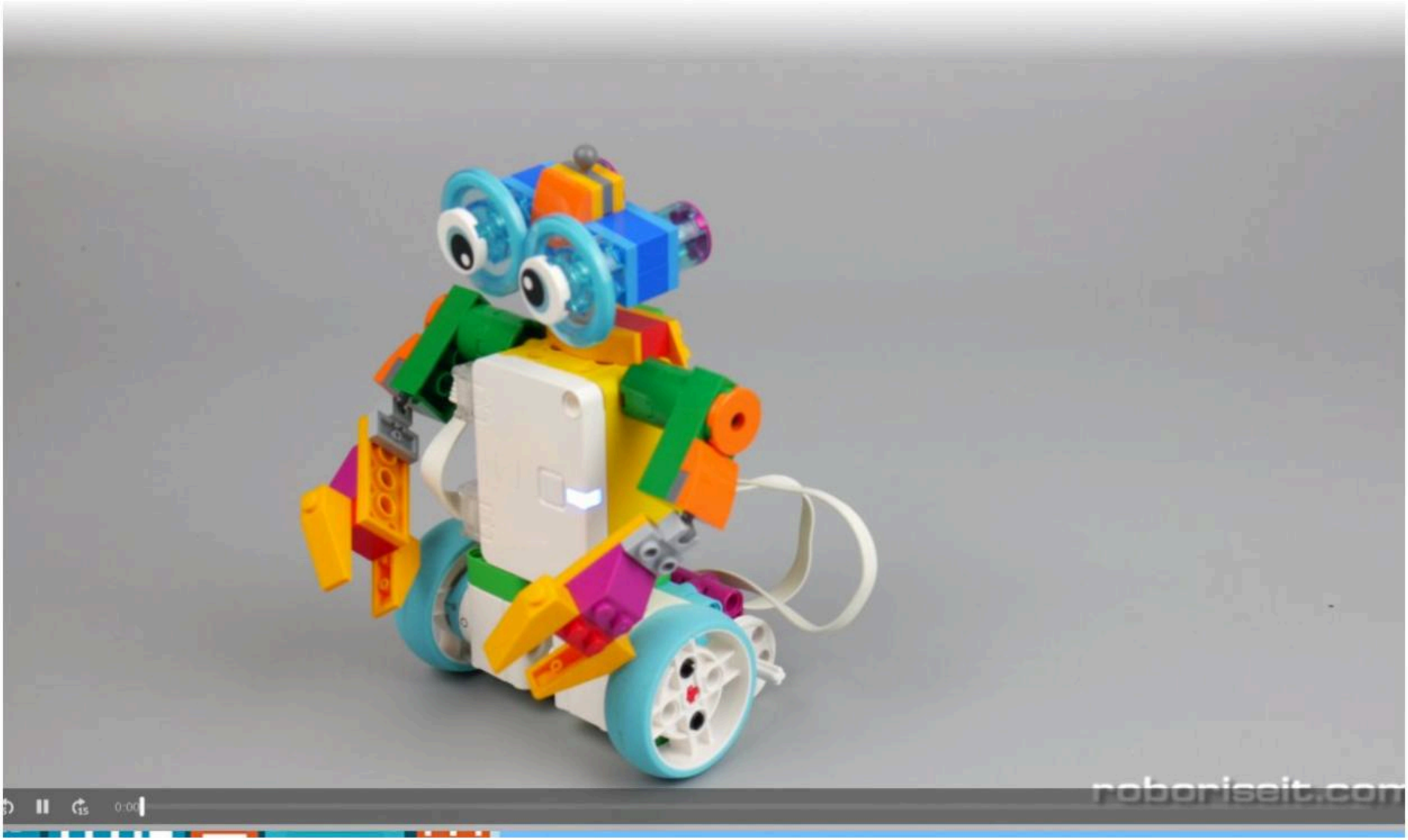


Repeat the robot test

Measure the distance traveled by the robot in one turn of the wheel. Thanks to this, you will be able to predict the block settings needed in the next steps

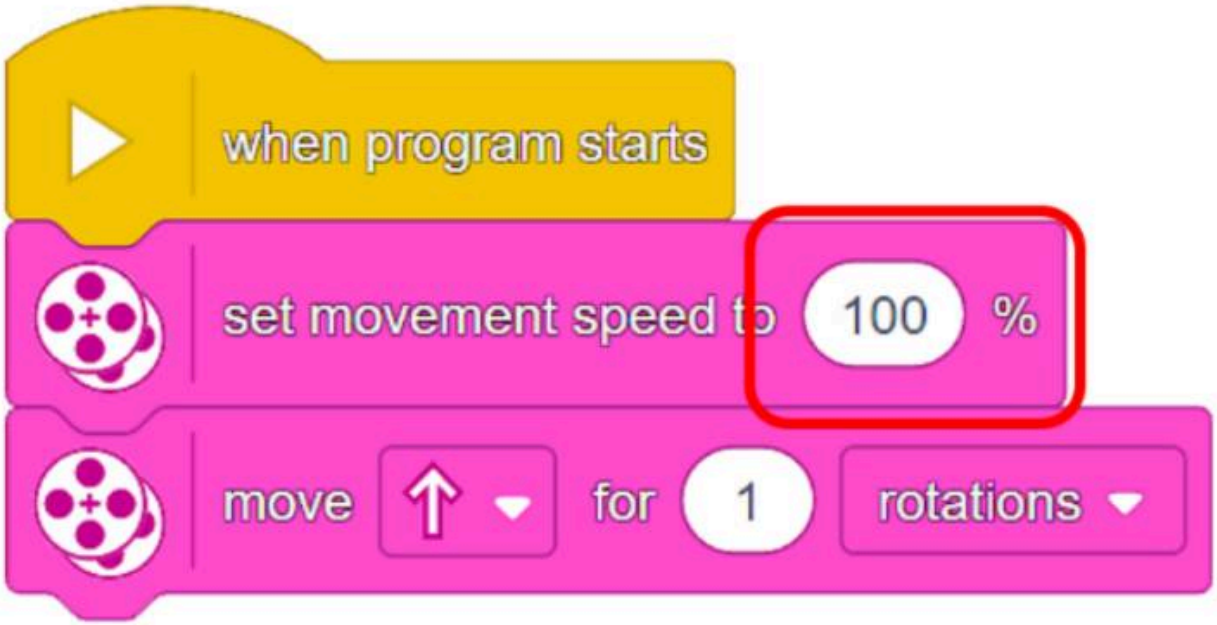


Experiment with the movement speed of your robot. Set the maximum speed and test the following program



Use the blocks to program the movement of the robot:

In Word Blocks mode, the program looks like this:



Repeat the robot test

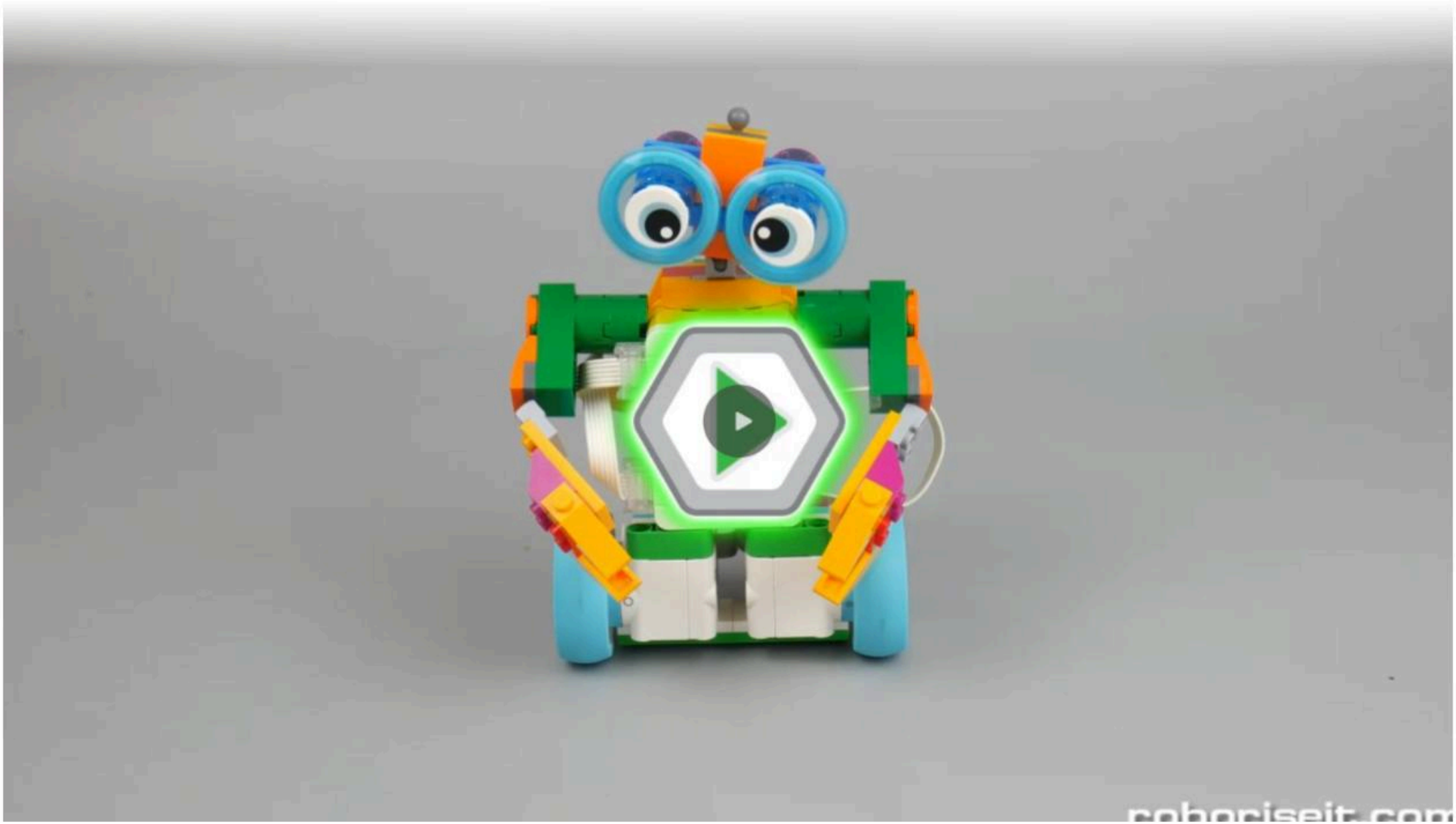
Use the blocks to program the movement of the robot:

In ICON Blocks mode, the program looks like this:



Repeat the robot test

The next movement your robot needs to learn is turning. Set the rotation duration of the motors to allow the robot to return exactly to the left or right



Note that it makes sense to use setting the rotation
the motor axis, rather than in turns, to simplify fine-tu

One turn consists of 360 degrees. Thus, turning the
turn. But 90 degrees of rotation of the robot is not 9
axis. And you program exactly the number of degree
Therefore, turning the robot to the left by 90 degree
degree rotation of the motor axis.

In ICON Blocks mode, the program looks like this:

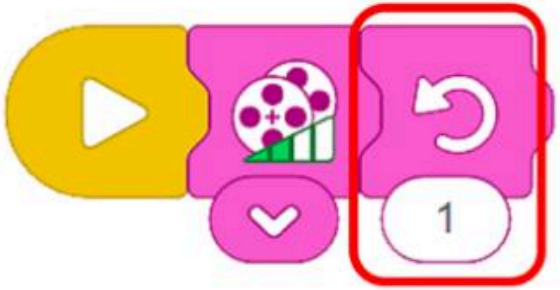


Note that it makes sense to use setting the rotation duration in degrees of rotation of the motor axis, rather than in turns, to simplify fine-tuning.

One turn consists of 360 degrees. Thus, turning the robot left or right is a 90-degree turn. But 90 degrees of rotation of the robot is not 90 degrees of rotation of the motor axis. And you program exactly the number of degrees by which the motor axis will turn. Therefore, turning the robot to the left by 90 degrees may require, for example, a 170-degree rotation of the motor axis.

In ICON Blocks mode, the program looks like this:





Note that it makes sense to use setting the rotation duration in degrees of rotation of the motor axis, rather than in turns, to simplify fine-tuning.

One turn consists of 360 degrees. Thus, turning the robot left or right is a 90-degree turn. But 90 degrees of rotation of the robot is not 90 degrees of rotation of the motor axis. And you program exactly the number of degrees by which the motor axis will turn. Therefore, turning the robot to the left by 90 degrees may require, for example, a 170-degree rotation of the motor axis.

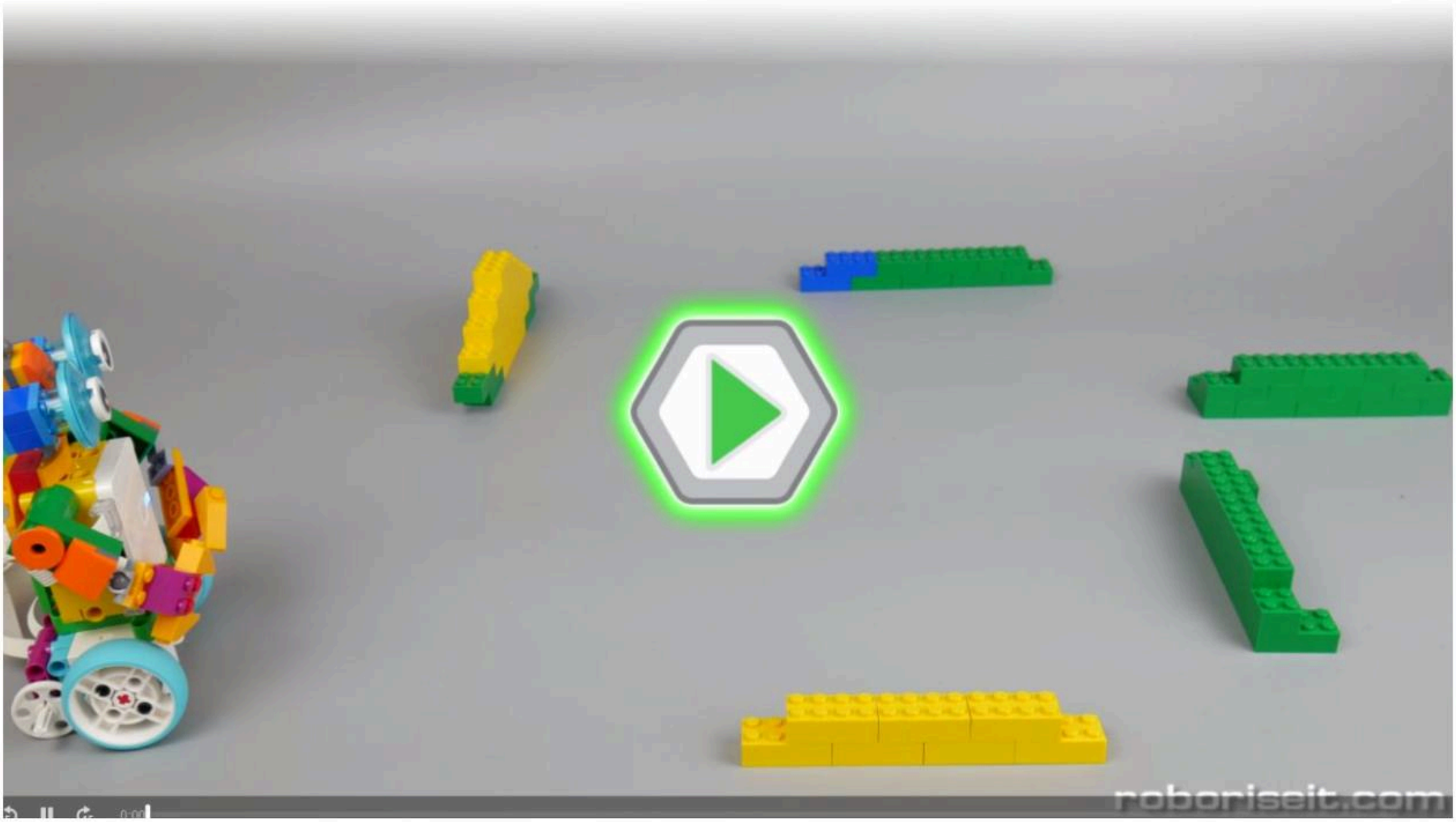
In Word Blocks mode, the program looks like this:



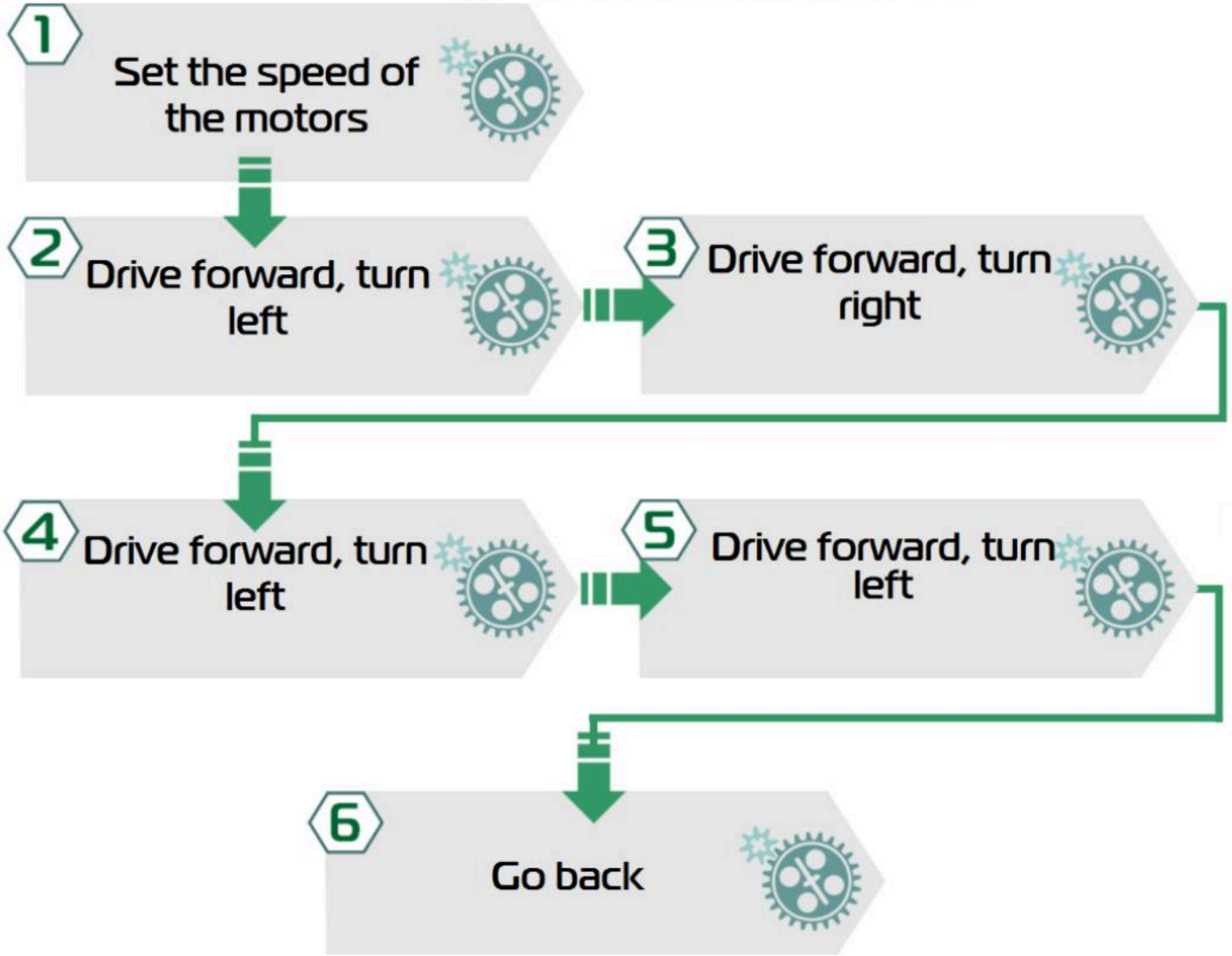


This setting changes the trajectory of the robot. At '0' the robot goes straight, and at '100' the wheels rotate in different directions at the same speed and the robot turns in place.

Now you are ready! Combine your programs so that the robot can cut the lawn in the yard.



The program should work according to the following algorithm:



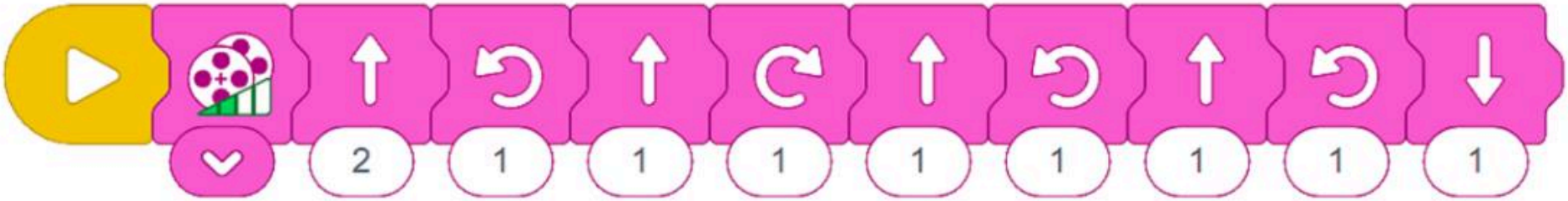
Program and test your robot.

In Word Blocks mode, the program looks like this:



Program and test your robot.

In ICON Blocks mode, the program looks like this:



What is the correct sequence of setting the motor's speed and duration?



speed first

duration first

programs will work equally

Select the robot's parts that make him move.



Discuss!

- What changes in nature prompted Sean to create home robots?
- What are the robos? What tasks do they solve?
- How does an algorithm differ from a program?
- Why did you have to use special blocks to program the forward movement of the robot?
- How many motors does your robot have?
- Why else would you want to train your robot?

